Bridging Low-Level and High-Level Visions: A Hybrid Transformer-CNN Multitask Framework for Hyperspectral Image Classification

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***Abstract*—Hyperspectral images (HSIs) are often subject to blurring during the imaging process, particularly around class boundaries. Such blurring typically results in the loss of spatial details, posing a great challenge to the HSI classification task, especially when the number of training samples is limited. Super-resolution (SR), as a powerful tool in low-level vision tasks, can recover fine-grained spatial details from degraded images. However, most existing methods for HSI classification and SR are implemented independently. On the one hand, this paradigm neglects the connection between low-level and high-level vision tasks; on the other hand, the multitasking framework enables a more comprehensive exploration of the feature space with limited training samples. In this article, we propose a novel multitask framework that integrates SR as an auxiliary task to enhance classification performance. Specifically, we exploit the strengths of a convolution neural network (CNN) and a transformer in extracting high-frequency and low-frequency information to construct a dual-branch network and design a frequency-domain interaction module (FIM) to facilitate information exchange between the two branches for a powerful representation capability. In addition, we propose a cross-task interaction module (CTIM) that effectively integrates spatial details captured by the SR task into the classification task. Furthermore, we propose a frequency-domain contrastive learning (FCL) to enforce the model to learn the consistency between both tasks and boost the instance discriminability. Extensive experiments on four hyperspectral datasets demonstrate that the proposed method achieves competitive results compared with the state-of-the-art (SOTA) HSI classification methods.**

***Index Terms*—Contrastive learning, convolution neural network (CNN), hyperspectral image (HSI) classification, HSI super-resolution (SR), multitask, transformer.**

Received 11 June 2025; revised 3 September 2025 and 12 October 2025; accepted 16 October 2025. Date of publication 20 October 2025; date of current version 30 October 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62036006 and Grant 62376205; and in part by the Fundamental Research Funds for the Central Universities, China, under Grant ZYTS25141. *(Corresponding authors: Mingyang Zhang; Maoguo Gong.)*

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Digital Object Identifier 10.1109/TGRS.2025.3623277

I. Introduction

HYPERSPECTRAL images (HSIs) contain tens or even hundreds of spectral bands at each pixel, which can offer much more abundant spatial and spectral information and reflect the subtle spectral properties of different objects [1], [2]. Consequently, HSIs have been applied to numerous applications, such as medical diagnosis [3], food quality [4], and plant monitoring [5], among others [6]. Over the past few decades, HSI classification has attracted much attention.

With the continuous advancement of HSI sensors, the spatial resolution of HSI has been progressively improved, which may introduce imaging blurring, especially for the inter-class boundary [7]. The performance of HSI classification is highly dependent on image quality. These blurry images pose a significant challenge to accurate classification, as the loss of spatial details can hinder the discrimination of fine-grained land-cover categories [8]. Super-resolution (SR), as a technique for enhancing the spatial resolution of images, aims to reconstruct high-resolution images from degraded low-resolution inputs. It can effectively sharpen blurry images and enrich spatial details. HSI SR is categorized into fusion-based HSI SR and single HSI SR. The fusion-based methods mainly use some auxiliary images with a larger spatial resolution, which limits their applicability in real-world scenarios [9]. In recent years, deep learning-based single HSI SR methods have developed rapidly [10]. Since HSIs generally contain dozens or even hundreds of spectral channels, the group-based methods have been extensively studied to reduce the model size [11], [12], [13]. For instance, SSPSR [14] adopted the grouping strategy and utilized a progressive multibranch network to learn a spatial-spectral prior. Considering the advantage of transformer in long-range dependency modeling, Liu et al. [15] proposed a hybrid architecture that combines 3-D CNN and transformer to jointly capture complementary information. Chen et al. [16] designed a cross-scope spatial-spectral transformer (CST) to adequately capture the long-range dependencies in both spatial and spectral dimensions. However, these methods primarily enhance the visual quality of images without imparting practical application significance, which means the potential of low-level vision tasks to boost high-level tasks, such as classification, remains underexplored.

Deep learning-based HSI classification methods have achieved impressive results. Compared to traditional machine

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learning approaches, they do not rely on handcrafted design and can extract deeper feature representations from high-dimensional HSI. The method based on CNN was initially applied to deep spectral feature extraction [17]. Subsequently, some researchers utilized CNN to extract spatial features [8]. Considering the three-dimensional characteristic of HSI data, methods based on 3-D CNN [18], [19] and 2-D-3-D CNN [20] were developed to explore joint spectral-spatial features, further improving classification performance. However, due to the limitation of the local receptive field provided by the convolution kernel, these methods cannot model global features. Recently, transformer has achieved great success in computer vision [21], leveraging its ability to model long-range dependencies. Meanwhile, a growing number of HSI classification methods based on transformer have also emerged [22]. For instance, Roy et al. [23] proposed the morphological transformer (MorphFormer) method that integrates morphological operations with self-attention mechanisms to effectively capture both local structural patterns and global spectral-spatial features. To reduce the learnable parameters in stacked transformer architectures, Zhao et al. [24] incorporated the groupwise separable strategy into the transformer to capture global spectral-spatial information. To efficiently capture both local and global features in HSIs, the hybrid CNN-transformer framework has garnered significant interest. By leveraging the complementary strengths of both architectures, these approaches demonstrated remarkable competitiveness [25], [26]. For instance, Huang et al. [27] designed the spatial-spectral transformer network with multiscale convolution (SS-TMNet) that combines a multiscale 3-D convolution projection module with the spatial-spectral attention module to extract global spatial-spectral features. Zhou et al. [28] proposed the dual-branch convolutional transformer network (DCTN) that aggregates local and global spatial-spectral features. However, these methods primarily exploit high-level semantic information beneficial for classification, without fully exploring the potential of extracting additional useful information from the redundant data inherent in HSIs, thereby limiting further improvements in classification performance.

Multitask learning (MTL) aims to jointly learn multiple related tasks and facilitate knowledge sharing between tasks to enhance the model's generalization ability [29]. In the field of HSI classification, some studies have focused on leveraging the multitask framework to address the inherent challenges of HSI data, aiming to enhance the classification performance. Considering the mixed pixel problem inherent in HSIs, some researchers integrated the unmixing task with the classification task to enhance the robustness and generalization capability in practical applications [30], [31]. Ibañez et al. [32] integrated the mask reconstruction with classification to mitigate the inherent noise issues in HSI data. To alleviate the issue of limited labeled samples and capture the rich information, some researchers have combined representation learning with HSI classification tasks, such as generative reconstruction [33], mask reconstruction [34], and contrastive learning [35], [36], among others. Although these methods efficiently explore HSI information from multiple perspectives, the inherent blurriness of HSI data leads to the loss of spatial detail, which hinders further improvement in performance.

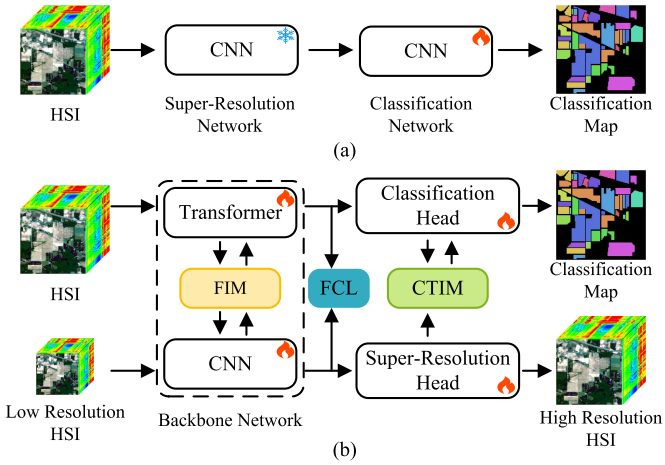


Fig. 1. Illustration to clarify the similarities and differences between the previous work [7] and ours. The snowflake denotes the module with pre-trained weights that are frozen, while the spark indicates the trainable module. (a) Workflow for the previous work [7]. (b) Workflow for the proposed method.

Motivated by these factors, we propose a multitask framework that integrates SR and classification, aiming to bridge low-level and high-level vision tasks, thereby endowing the low-level vision task with practical application significance. The work most similar to ours is [7]. The SR network is pre-trained as a method to enhance the input image, followed by training the classification network. However, this approach has some limitations: on one hand, the two-stage training process is relatively complex; on the other hand, the pre-trained SR network primarily enhances the image visually, without fully exploring the deep connections between SR and classification. When humans interpret a blurry image, a common strategy is to gradually zoomed-in view on the image while making visual judgments, which often leads to more reliable outcomes. Inspired by this behavior, our approach integrates SR and classification tasks within a unified framework. By enriching image details through the SR task and simultaneously transmitting this information to the classification network, our method facilitates more accurate and robust classification decisions.

Fig. 1 briefly illustrates the similarities and differences between the previous work and ours. Different from the work [7] that performs SR and classification sequentially, we integrate both tasks into a multitask framework, optimizing them simultaneously without any pre-training to capture richer, detailed information and thereby improve the classification performance. We design a powerful backbone network based on a hybrid CNN-transformer architecture, which has been proven to be effective in the field of HSI classification [25], [26], [27], [28]. Simply concatenating them cannot fully leverage the complementary advantages of both architectures. Inspired by [37] and [38], CNN can be regarded as a high-pass filter to extract high-frequency information, and transformer can be regarded as a low-pass filter to extract low-frequency information. Therefore, we propose a frequency-domain interaction module (FIM) to facilitate information exchange for

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a more powerful representation capability. Moreover, it has been demonstrated that information inside one task can assist in the prediction of other tasks [39]. Simply using a unified backbone network to learn a shared representation overlooks the relationships between tasks. Therefore, we design a cross-task interaction module (CTIM) that leverages internal information from the SR task to assist the classification task. To enhance the instance discrimination ability, considering the characteristics of both SR and classification tasks, we design a frequency-domain contrastive learning (FCL), which alleviates the issue of limited training samples to some extent. Furthermore, this similarity learning helps align the feature spaces of the SR and classification tasks, improving the representational capacity of our method.

In summary, the contributions of our work can be summarized as follows.

1) In this work, we integrate HSI SR and classification into a multitask framework to bridge low-level and high-level vision tasks, thereby fully exploring the feature space with limited labeled samples.
2) We leverage the strengths of CNN and transformer in capturing high-frequency and low-frequency information to design a dual-branch network architecture. In addition, we propose an FIM to facilitate information exchange for a more powerful representation capability.
3) We propose a CTIM that aggregates the information from SR into the classification head from a frequency-domain perspective, enhancing classification performance.
4) We introduce an FCL that uses the original images and low-resolution counterparts to encourage the model to learn the consistency between SR and classification and bring the instance discriminability.

The structure of this article is constructed as follows. Section II briefly reviews some related works. In Section III, the proposed method is described in detail. Section IV reports the comparison and the ablation experimental results. Conclusions are given in Section V.

II. Preliminaries and Motivation

A. Super-Resolution

Image SR, as an important class of image processing techniques in computer vision, aims to recover the corresponding high-resolution images from the low-resolution images. Generally, the low-resolution image I_{LR} is modeled as the output of the following degradation:

$$I_{LR} = \mathcal{D}(I_{HR}; \delta) \tag{1}$$

where $\mathcal{D}$ denotes a degradation mapping function, I_{HR} is the corresponding high-resolution image, and δ is the parameters of the degradation process (e.g., the scaling factor or noise). Most works directly model the degradation as a single downsampling operation, as follows:

$$\mathcal{D}(I_{HR}; \delta) = (I_{HR}) \downarrow_s, \quad \{s\} \subset \delta \tag{2}$$

where $\downarrow_s$ is a downsampling operation with the scaling factor s. Meanwhile, some works [40] model the degradation process as a combination of several operations

$$\mathcal{D}(I_{HR}; \delta) = (I_{HR} \otimes \kappa) \downarrow_s + n_\zeta, \quad \{s, \kappa, \zeta\} \subset \delta \tag{3}$$

where $I_{HR} \otimes \kappa$ represents the convolution between a blur kernel κ and n_ζ is some additive white Gaussian noise with standard deviation ζ. Compared to (2), the combinative degradation pattern of (3) is closer to real-world cases.

Based on the above, researchers have begun exploring a variety of methods to achieve SR, following:

$$I_{SR} = \mathcal{F}(I_{LR}; \theta) \tag{4}$$

where $\mathcal{F}$ is the SR model and θ denotes the parameters of $\mathcal{F}$. The objective of SR is as follows:

$$\widehat{\theta} = \arg\min_x \mathcal{L}(I_{SR}, I_{HR}) + \lambda \Phi(\theta) \tag{5}$$

where $\mathcal{L}(I_{SR}, I_{HR})$ represents the loss function between the SR image I_{SR} and the ground truth I_{HR}, $\Phi(\theta)$ is the regularization term, and λ is the tradeoff parameter. The most popular loss function for SR is ℓ_1 loss.

With the development of deep learning, a large number of HSI SR works have emerged, including methods based on CNN [41], transformer [15], [42], among others [10]. Considering the high-dimensional spectral characteristics of HSIs, directly reconstructing all channels requires a large model. Therefore, some researchers have started to pay attention to group-based methods, which can reduce model size while still achieving promising results [12], [14]. However, existing methods mainly focus on improving image reconstruction quality, while overlooking a key question: what is the purpose or significance of enhancing image quality? Inspired by this, we aim to bridge low-level and high-level vision by exploring how knowledge from low-level vision tasks can be leveraged to support and enhance high-level vision tasks.

B. Multitask Learning

Focusing solely on a single task may result in limited feature representation, potentially overlooking other valuable information. In contrast, humans often leverage knowledge acquired from one task to facilitate learning in another, demonstrating the effectiveness of shared learning across tasks. Inspired by this, MTL, a learning paradigm in machine learning, simultaneously optimizes multiple related tasks, which enables knowledge transfer across tasks, thereby enhancing the generalization capability of all tasks [29]. Given m related tasks $\{\mathcal{T}_i\}_{i=1}^m$, MTL aims to jointly learn these tasks to enhance the performance of each individual task $\mathcal{T}_i$ by leveraging the knowledge contained in the other tasks. Each task $\mathcal{T}_i$ contains a training dataset $\mathcal{D}_i$ consisting of n_i training samples, i.e., $\mathcal{D}_i = \{\boldsymbol{x}_j^i, y_j^i\}_{j=1}^{n_i}$, where $\boldsymbol{x}_j^i$ is the jth training instance and y_j^i is its label.

With the rapid development of MTL in the field of computer vision, researchers have been motivated to introduce it into HSI processing, including the integration of auxiliary tasks with classification tasks. For instance, Ibañez et al. [32] proposed the MAEST that integrates masked reconstruction

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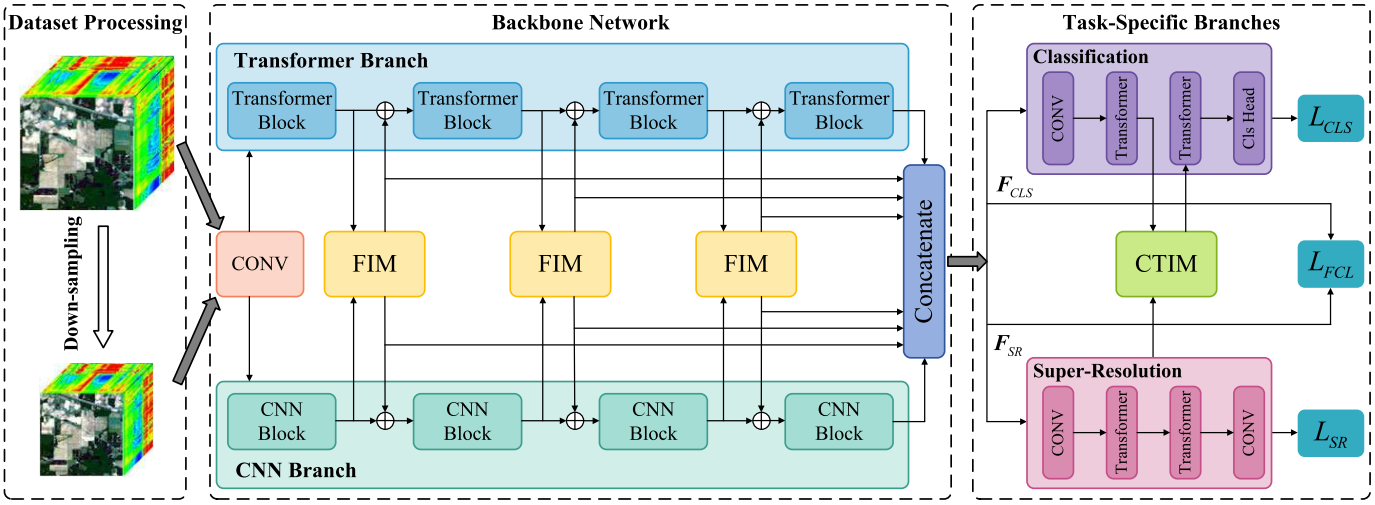


Fig. 2. Overall architecture of our method, in which "CONV" refers to the convolutional layer, "FIM" refers to the frequency-domain interaction module, "CTIM" refers to the cross-task interaction module, "Cls Head" refers to the predictor of the classification branch including an adaptive average pooling layer and a fully-connection layer, "L_{FCL}" refers to the loss function of the FCL, "L_{CLS}" denotes the loss function of classification, and "L_{SR}" denotes the loss function of SR.

with classification to mitigate the inherent noise issues in HSIs. Han et al. [31] combined the unmixing task with the classification task to alleviate the mixed pixel problem in HSIs, thereby enhancing the robustness and generalization in practical applications. Considering the limited labeled samples, Hang et al. [33] proposed a multitask generative adversarial network (MTGAN) that combined the reconstruction and classification tasks to alleviate the requirement for training samples. Lu et al. [35] integrated contrastive learning and classification into an end-to-end framework, which enables the model to learn both the similarity and dissimilarity of sample pairs with limited labeled samples. Huang et al. [36] integrated masked reconstruction, contrastive learning, and classification into a unified framework. The masked reconstruction task introduces local inductive bias, while the contrastive learning between masked and unmasked views enhances instance discriminability of the network.

These studies demonstrate the potential of MTL in HSI tasks. However, they overlook a critical issue: the inherent blurriness in HSI data [7]. Accurate classification heavily depends on image quality, and the blurry data poses challenges for precise recognition. SR, on the other hand, can enrich image details and sharpen blurred images. However, simply cascading the two tasks [7] fails to fully exploit the inter-task relationships, making it difficult to unleash the potential benefits of SR for classification. Inspired by this, we explore integrating SR and classification into a unified multitask framework to capture more generalizable features and improve classification accuracy.

III. Proposed Method

A. Overall Framework

The overall framework of our method is shown in Fig. 2, which consists of three key components: the dataset processing, the backbone network, and the task-specific branches. In the dataset processing stage, the original input images used for classification are downsampled to generate low-resolution counterparts, which serve as inputs for the SR task. Then, the input images and their low-resolution counterparts are fed into the shared backbone network for feature extraction. In the backbone network, a convolutional layer is first employed to project the inputs with varying spectral dimensions into a unified channel dimension. Subsequently, a dual-branch network architecture based on transformer [43] and CNN [44] is designed to extract global and local features, respectively. To fully exploit their respective strengths in capturing low-frequency and high-frequency information, a FIM is introduced, further enhancing the representational capability of the network. Compared with the previous work based on CNN [7], our hybrid architecture not only effectively captures local information but also models long-range dependencies, thereby achieving a more powerful representation capability. Finally, the output features are concatenated to form a unified representation. The task-specific branches consist of three components: the classification branch, the SR branch, and the FCL branch, which enhances the instance discrimination by leveraging the original images and low-resolution counterparts to generate plausible contrastive views. Among them, the CTIM is employed to aggregate SR features into the classification task. Unlike direct concatenation [7], this interaction module effectively establishes cross-task connections, enabling the detailed features captured by the SR task to better assist the classification task. Through leveraging three distinct task-specific branches, our method enables comprehensive exploration of the feature space from multiple perspectives, thereby increasing the amount of learning information with limited training samples. It is worth noting that during the training phase, we use the original images and low-resolution counterparts as inputs for the classification and SR tasks, respectively. This aims to establish a connection between the two tasks. During the test stage, the SR is performed on the original images to enrich spatial details, thereby facilitating more accurate classification.

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B. SR Branch

SR, as a powerful tool in low-level vision, reconstructs high-resolution images from degraded ones by enriching image details. In the proposed method, we employ SR as an auxiliary task to provide rich, detailed features for the classification task, thereby enhancing classification performance, as shown in Fig. 2.

Let $\boldsymbol{I} \in \mathbb{R}^{B\times H\times W}$ denote the input HSI cube, where B is the number of spectral bands and $H \times W$ is the spatial size. In the data processing stage, to construct data pairs for the SR task, we generate degraded versions of the original input images by applying the bilinear downsampling. The low-resolution data is expressed as $\boldsymbol{I}_{LR} \in \mathbb{R}^{B\times H/s\times W/s}$, where s is the scale factor. Since the classification and SR tasks share the backbone network, we address the mismatch in feature space dimensions by applying bilinear upsampling to the input of the SR task to ensure that the intermediate feature maps of both tasks have the same spatial dimensions. Therefore, the input of the SR task is the upsampled version of $\boldsymbol{I}_{LR}$, which is expressed as $\boldsymbol{I}_{UP} \in \mathbb{R}^{B\times H\times W}$.

Then, $\boldsymbol{I}_{UP}$ is fed into the backbone network. To accommodate the SR task and minimize information loss during feature extraction, our backbone network does not perform feature compression, maintaining a consistent spatial resolution throughout the entire process. Through the backbone network, we obtain the feature of SR, which can be denoted as

$$\boldsymbol{F}_{SR} = f_{\text{backbone}}\left(\boldsymbol{I}_{UP}\right) \tag{6}$$

where $\boldsymbol{F}_{SR} \in \mathbb{R}^{C\times H\times W}$ is the feature of the SR task and f_{backbone} is the backbone network. C denotes the channel dimension, which is demonstrated in the implementation details. The feature $\boldsymbol{F}_{SR}$ is then fed into the SR head to generate the high-resolution output $\boldsymbol{I}_{SR} \in \mathbb{R}^{C\times H\times W}$, which can be denoted as

$$\boldsymbol{I}_{SR} = f_{SR_\text{head}}\left(\boldsymbol{F}_{SR}\right) \tag{7}$$

where f_{SR_head} is the SR head network.

It is worth noting that we leverage the feature within the SR head for interaction with the classification task, serving as one of the inputs to the CTIM module. Meanwhile, the output feature of the backbone network $\boldsymbol{F}_{SR}$ is used as an input to the FCL module. The reasons for this design are as follows. The backbone network, as a feature extractor, focuses on capturing both global and local features efficiently. Performing contrastive learning on features from different tasks enhances the representational capacity of the backbone network. In the SR head, the feature is modulated specifically for generating high-resolution images, thus containing richer information and more details, which is more beneficial for the classification task.

C. FIM Module

The transformer architecture is capable of modeling global relationships and long-range dependencies through its self-attention mechanism, effectively capturing the semantic and structural information of the input image [37]. This self-attention mechanism averages feature maps, reducing the variance of feature map points, and thus tends to act as a low-pass filter that reduces high-frequency signals [38]. However, due to the constraint of the sliding window, CNN can directly capture pixel-level variations, making it pay more attention to local features, such as texture and details. So, CNN can be regarded as a high-pass filter, which is more suitable for extracting high-frequency information [37]. Therefore, to some extent, transformer and CNN are complementary [38]. Inspired by this, we design the FIM module.

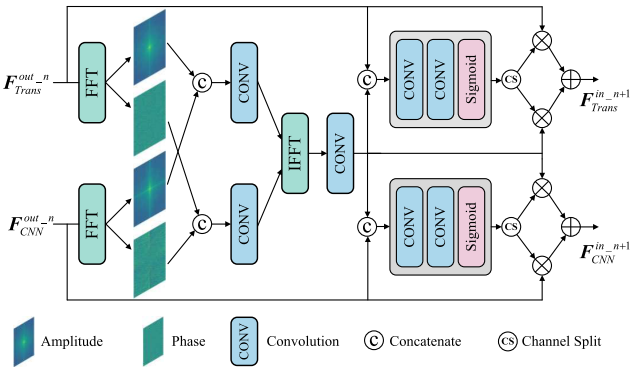


Fig. 3. Structure of the FIM module, in which "FFT" refers to the fast Fourier transform and "IFFT" refers to the inverse FFT.

Fig. 3 illustrates the FIM module proposed in our method, which leverages the respective advantages of transformer and CNN from a frequency-domain perspective to facilitate the information interaction for a more powerful representation capability. Let $\boldsymbol{F}_{\text{Trans}}^{\text{out}_n}$ with a size of $C \times H \times W$ denote the output of the nth transformer block and $\boldsymbol{F}_{\text{CNN}}^{\text{out}_n}$ with a size of $C \times H \times W$ denote the output of the nth CNN block, where C denotes the channel dimension, and H and W represents the height and width dimensions, respectively. First, both features are transformed into the frequency domain using the fast Fourier transform (FFT), and their corresponding amplitude and phase components are concatenated, respectively. The process can be denoted as

$$\begin{aligned}\boldsymbol{F}_{\text{Trans}}^{\text{Amp}}, \boldsymbol{F}_{\text{Trans}}^{\text{Pha}} &= \text{FFT}\left(\boldsymbol{F}_{\text{Trans}}^{\text{out}_n}\right)\\ \boldsymbol{F}_{\text{CNN}}^{\text{Amp}}, \boldsymbol{F}_{\text{CNN}}^{\text{Pha}} &= \text{FFT}\left(\boldsymbol{F}_{\text{CNN}}^{\text{out}_n}\right).\end{aligned} \tag{8}$$

Then, these concatenated representations are passed through a convolutional layer for feature fusion and dimensionality reduction, yielding the fused amplitude and phase. It can be formulated as

$$\begin{aligned}\boldsymbol{F}_{\text{fuse}}^{\text{Amp}} &= H_1\left(\left[\boldsymbol{F}_{\text{Trans}}^{\text{Amp}}, \boldsymbol{F}_{\text{CNN}}^{\text{Amp}}\right]\right)\\ \boldsymbol{F}_{\text{fuse}}^{\text{Pha}} &= H_2\left(\left[\boldsymbol{F}_{\text{Trans}}^{\text{Pha}}, \boldsymbol{F}_{\text{CNN}}^{\text{Pha}}\right]\right)\end{aligned} \tag{9}$$

where $H(\cdot)$ denotes the 3×3 convolution layer. The fused frequency-domain features are transformed back into the spatial domain using the inverse FFT (IFFT), which can be formulated as

$$\boldsymbol{F}_{\text{fuse}} = \text{IFFT}\left(\boldsymbol{F}_{\text{fuse}}^{\text{Amp}}, \boldsymbol{F}_{\text{fuse}}^{\text{Pha}}\right) \tag{10}$$

where $\boldsymbol{F}_{\text{fuse}}$ denotes the fused features with a size of $C\times H\times W$. Since the fused information may still contain redundancy or suffer from information loss, we propose to predict the pixel-wise attention to adaptively integrate the fused feature with the original transformer and CNN features. Taking the former as an example, we concatenate the fused feature and the transformer feature and feed the concatenated result into

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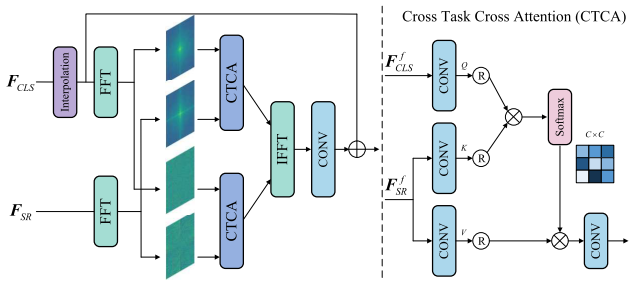


Fig. 4. Structure of the CTIM module, in which "R" refer to the resize.

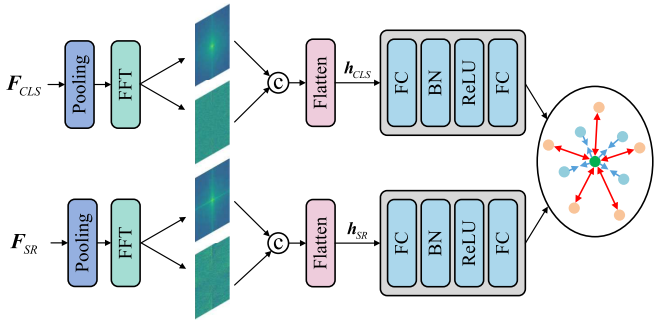


Fig. 5. Structure of the FCL module, in which "Pooling" refers to the adaptive average pooling layer, "FC" refers to the fully-connected layer, and "BN" refers to the batch normalization. The blue dashed lines mean pulling toward each other, whereas the red dashed lines mean pushing away from each other.

the pixel-wise attention network f_{PA} that consists of two 3×3 convolutional layers and a sigmoid function. It can be formulated as

$$\boldsymbol{F}_{\mathrm{Att}} = f_{PA}\left(\left[\boldsymbol{F}_{\mathrm{fuse}}, \boldsymbol{F}_{\mathrm{Trans}}^{\mathrm{out}_n}\right]\right). \tag{11}$$

We split the $\boldsymbol{F}_{\mathrm{Att}}$ along the channel dimension to generate adaptive attentions $\boldsymbol{F}_{\mathrm{Trans}}^{\mathrm{Att}}$ and $\boldsymbol{F}_{\mathrm{fuse}}^{\mathrm{Att}}$ for reweighting them adaptively

$$\boldsymbol{F}_{\mathrm{Trans}}^{\mathrm{in}_n+1} = \boldsymbol{F}_{\mathrm{Trans}}^{\mathrm{Att}} \times \boldsymbol{F}_{\mathrm{Trans}}^{\mathrm{out}_n} + \boldsymbol{F}_{\mathrm{fuse}}^{\mathrm{Att}} \times \boldsymbol{F}_{\mathrm{fuse}} \tag{12}$$

where $\boldsymbol{F}_{\mathrm{Trans}}^{\mathrm{in}_n+1}$ denotes the input of the $n+1$th transformer block.

D. CTIM Module

To leverage the internal information of the SR task to assist the classification task, we design a CTIM, as illustrated in Fig. 4. Since the SR task takes low-resolution images as input and reconstructs high-frequency details through a reconstruction network, we establish a frequency-domain information transfer pathway to bridge both tasks. Considering the spatial dimension difference between SR features and classification features, we adopt the bilinear interpolation to the classification features to ensure both features have the same spatial size. Let $\boldsymbol{F}_{\mathrm{CLS}}$ with a size of $C \times H \times W$ denote the feature of the classification task and $\boldsymbol{F}_{SR}$ with a size of $C \times H \times W$ denote the feature of the SR task. First, we apply FFT to obtain the amplitudes ($\boldsymbol{F}_{\mathrm{CLS}}^{\mathrm{Amp}}, \boldsymbol{F}_{SR}^{\mathrm{Amp}} \in \mathbb{R}^{C\times H\times(W/2+1)}$) and phases ($\boldsymbol{F}_{\mathrm{CLS}}^{\mathrm{Pha}}, \boldsymbol{F}_{SR}^{\mathrm{Pha}} \in \mathbb{R}^{C\times H\times(W/2+1)}$). Then, these features are fused by the cross-task cross-attention (CTCA) mechanism.

To be specific, CTCA contains two inputs: $\boldsymbol{F}_{\mathrm{CLS}}^{f}$ and $\boldsymbol{F}_{SR}^{f}$, which f denotes the frequency feature (i.e., amplitude or phase). $\boldsymbol{F}_{\mathrm{CLS}}^{f}$ is used to generate query ($\boldsymbol{Q}$), and $\boldsymbol{F}_{SR}^{f}$ is used to obtain the key ($\boldsymbol{K}$) and value ($\boldsymbol{V}$) by applying different convolutional layers. It can be denoted as

$$\boldsymbol{Q} = W_q\left(\boldsymbol{F}_{\mathrm{CLS}}^{f}\right), \boldsymbol{K} = W_k\left(\boldsymbol{F}_{SR}^{f}\right), \boldsymbol{V} = W_v\left(\boldsymbol{F}_{SR}^{f}\right) \tag{13}$$

where $W(\cdot)$ denotes the function of convolution. So, we have $\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V} \in \mathbb{R}^{C\times H(W/2+1)}$. After that, the cross-attention matrix ($\boldsymbol{M} \in \mathbb{R}^{C\times C}$) is calculated in the spectral dimension to decrease the computational cost. It can be formulated as

$$\begin{aligned} \boldsymbol{M} &= f_{\mathrm{softmax}}\left(\boldsymbol{Q}\boldsymbol{K}^T/\alpha\right) \\ \boldsymbol{F}_{\mathrm{attn}} &= H_3\left(\boldsymbol{M}\boldsymbol{V}\right) \end{aligned} \tag{14}$$

where α is a learnable scaling parameter to control the magnitude of the dot product of $\boldsymbol{Q}$ and $\boldsymbol{K}$ before applying the softmax function, H_3 denotes the convolutional layer, and $\boldsymbol{F}_{\mathrm{attn}}$ is the result. The fused amplitude and phase components are transformed back into the spatial domain via IFFT to obtain the cross-task fused features $\boldsymbol{F}_{\mathrm{cross}}$. Finally, the local residual learning is introduced to obtain the enriched feature, which can be formulated as

$$\boldsymbol{F}_{\mathrm{out}} = \boldsymbol{F}_{\mathrm{CLS}} + H_4\left(\boldsymbol{F}_{\mathrm{cross}}\right) \tag{15}$$

where H_4 denotes the convolutional layer and $\boldsymbol{F}_{\mathrm{out}}$ is result. Through this module, the frequency-domain correlations between the SR and classification tasks can be effectively learned, thereby leading to improved performance.

E. FCL Module

Contrastive learning performs representation learning by maximizing the consistency between views of the same sample in the latent space while increasing the disparity between different samples. Therefore, we integrate contrastive learning into our framework to enhance the instance discriminability and improve the overall representation capability. Considering our multitask framework, it is convenient and intuitive to take the original sample and the low-resolution sample as two views for the given training sample. The primary distinction between the views lies in the high-frequency components, as low-resolution images often lack edge and fine-grained details. Therefore, we propose an FCL strategy, as shown in Fig. 5.

To decrease the memory overheads, we first employ an adaptive average pooling layer to reduce the spatial dimensions. Then, we perform FFT to extract the amplitude and phase components, which are concatenated and flattened to construct representations ($\boldsymbol{h}_{\mathrm{CLS}}$ and $\boldsymbol{h}_{SR}$). For the FCL task, each of the two representations is fed into a corresponding projection head, which is made up of two fully-connected layers, a batch normalization, and a ReLU activation function. And the outputs are denoted z_{CLS} and z_{SR}, which are leveraged to formulate the FCL loss for model updating. The loss function adopts the form of InfoNCE [45]. Suppose that there are N training samples in in one mini-batch and $\{z_i\}_{i=1}^{2N}$ is used to denote all the representations for N samples

$$\begin{cases} z_{2k} = z_{\mathrm{CLS}}^{k} \\ z_{2k-1} = z_{SR}^{k} \end{cases} \tag{16}$$

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where $k = 1, 2, \ldots, N$, and z^k_{CLS} and z^k_{SR} are the representations of two views for the kth samples. Let $\mathrm{sim}(\boldsymbol{a}, \boldsymbol{b}) = \boldsymbol{a}^T\boldsymbol{b}/(\|\boldsymbol{a}\|\|\boldsymbol{b}\|)$ represent the inner product between ℓ_2 normalized $\boldsymbol{a}$ and $\boldsymbol{b}$. The contrastive loss function is defined as

$$\ell(i,j) = -\log \frac{\exp\left(\mathrm{sim}\left(z_i, z_j\right)\right)}{\sum_{k=1}^{2N} \mathbb{1}_{[k\neq i]} \exp\left(\mathrm{sim}\left(z_i, z_k\right)\right)}$$
$$\mathcal{L}_{\mathrm{FCL}} = \frac{1}{2N}\sum_{k=1}^{N}\left(\ell(2k-1, 2k) + \ell(2k, 2k-1)\right) \tag{17}$$

where $\mathbb{1}_{[\cdot]} \in \{0, 1\}$ is an indicator function returning 1 if and only if the condition in $[\cdot]$ is satisfied.

Through incorporating this unsupervised representation learning strategy, our method can capture the similarity and dissimilarity distribution properties of sample pairs in a feature representation space with limited training samples.

F. Loss Function

The overall loss of our proposed method consists of three components: classification loss, SR loss, and FCL loss.

The training of the classification task adopts the cross-entropy between the prediction and the ground truth. It can be formulated as

$$\mathcal{L}_{\mathrm{CLS}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C}\left(y_i^j \log\left(p_i^j\right)\right) \tag{18}$$

where C is the total number of land-cover classes. For y_i^j, when the label of the input x_i is equal to j, y_i^j is set to 1; otherwise, y_i^j is set to 0. For p_i^j, it denotes the probability that x_i is predicted to be jth class through our method.

For the SR task, we adopt ℓ_1 loss to measure the accuracy of the reconstructed images. The ℓ_1 loss is given as follows:

$$\mathcal{L}_{SR} = \frac{1}{N}\sum_{n=1}^{N}\left|\left|\boldsymbol{I}^n_{SR} - \boldsymbol{I}^n_{HR}\right|\right|_1 \tag{19}$$

where $\boldsymbol{I}^n_{SR}$ and $\boldsymbol{I}^n_{HR}$ refer to the nth reconstructed HSI and the corresponding ground truth.

The loss function of FCL is (17). Therefore, the total loss function is defined as a weighted sum of these three loss functions

$$\mathcal{L} = \mathcal{L}_{\mathrm{CLS}} + \lambda_1\mathcal{L}_{SR} + \lambda_2\mathcal{L}_{\mathrm{FCL}} \tag{20}$$

where λ_1 and λ_2 control the weights of different losses.

IV. Experiments and Results

We conduct extensive experiments to evaluate the performance of the proposed method in this section, structured as follows. Section IV-A describes four widely used HSI datasets for assessing all methods. Section IV-B details the experimental setup, including evaluation metrics and some state-of-the-art (SOTA) methods. Section IV-C describes the implementation details. Section IV-D compares the classification performance of our method with SOTA methods from quantitative and visual results. Section IV-E constructs ablation experiments to demonstrate the effectiveness of the proposed modules. Section IV-F analyzes the model performance under different settings of parameters, including the patch size of the input cube, the number of labeled samples, and combinations of loss weights. Finally, Section IV-G analyzes the computational complexity of the proposed method.

A. Datasets

1) Indian Pines Dataset: The Indian Pines dataset was collected by the NASA AVIRIS sensor over the agricultural Indian Pines test site in northwestern Indiana, USA, in 1992. The data cube has a spatial size of 145 × 145 pixels with a spatial resolution of 20 m and initially contains 220 spectral bands spanning the 0.4–2.5-μm range. After removing 20 bands severely affected by water absorption, 200 spectral bands are typically retained for further analysis. The ground truth contains 16 vegetation classes with 10 249 labeled pixels.

2) Salinas Dataset: The Salinas dataset was acquired by the AVIRIS imaging spectrometer over the Salinas Valley, California, USA. It has a spatial resolution of 3.7 m and comprises 512 × 217 pixels. After removing 20 water absorption bands, 204 spectral bands remain. The dataset is annotated with ground-truth labels for 16 distinct land-cover classes.

3) Kennedy Space Center Dataset: The Kennedy Space Center (KSC) dataset was collected by NASA AVIRIS over the Kennedy Space Center, Florida, USA, on March 23, 1996. The dataset comprises 224 spectral bands covering the 0.4–2.5-μm range with a spectral resolution of 10 nm, and has a spatial resolution of 18 m per pixel. The image consists of 512 × 614 pixels and includes ground-truth annotations for 13 upland and wetland land-cover classes, totaling 5211 labeled pixels. After removing 48 spectral bands affected by water absorption and low signal-to-noise ratio, 176 bands are retained for classification.

4) WHU-Hi-HanChuan Dataset: The WHU-Hi-HanChuan dataset was collected on June 17, 2016, in Hanchuan, Hubei, China, using a Headwall Nano-Hyperspec imaging sensor with a 17 mm focal length mounted on a Leica Aibot X6 UAV V1 platform. The dataset comprises 1217 × 303 pixels with a spatial resolution of 0.109 m and contains 274 spectral bands spanning the 0.4–1.0-μm wavelength range. It is annotated with 16 land-cover categories.

B. Experimental Setup

1) Evaluation Metrics: We employ three widely used evaluation metrics to quantitatively assess the performance of the proposed method and compare it with SOTA methods: overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa). Specifically, the OA is calculated as the ratio of correctly classified test samples to the total number of test samples. The AA is the mean of the classification accuracies across all classes. The Kappa is computed by weighting the measured accuracies, which represents the robust measure of the degree of agreement. Higher values of each metric indicate better classification performance. In addition, we visualized the classification results of the three datasets.

2) Compared Methods: To further highlight the advantages of our method, we conduct a comparative evaluation against SOTA methods, including the CNN and transformer architectures. Detailed descriptions of these methods are provided

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below. It is worth noting that all methods are compared using the experimental settings reported in their articles or re-implemented by their official code.

1) *M3D-DCNN [18]:* This model consists of a total of ten 3-D convolution layers and one fully connected layer. It is utilized to jointly learn both multiscale spatial features and spectral features from HSI data. This model is trained by the AdaGrad algorithm, in which the base learning rate is 0.01. In addition, the training epoch is set to 100, and the batch size is set to 40, weight decay of 0.01 for all the layers.
2) *3-D-CNN [19]:* This model consists of four 3-D convolutional layers, two 3-D pooling layers, a dropout layer, and one fully connected layer. It enables a joint spectral and spatial information process. The initial learning rate is 0.001, and it is divided by ten after 100 and 200 epochs. The number of training epochs is set to 300, and the batch size is set to 3.
3) *DFFN [46]:* This model constructs three levels of residual blocks and extracts more discriminative features by fusing multilevel outputs. The initial learning rate is 0.1 and then divided by 10 when the error plateaus. These networks are trained for 2×10^4 iterations, and the training minibatch has a size of 100.
4) *RSSAN [47]:* This model integrates attention mechanisms with residual modules. It employs a spectral attention mechanism to perform spectral band selection, and a spatial attention mechanism to adaptively extract spatial information relevant to classification. Optimization is performed by the RMSProp optimizer with a batch size of 16. The number of training epochs is set to 200, and the learning rate is set to 0.0003.
5) *SpectralFormer [22]:* This model includes a layer for processing spectral information in the neighbor band, a patch embedding layer, a transformer layer, and one fully connected layer for classification. This model adopts the Adam optimizer with a batch size of 64. The learning rate is initialized with 0.0005 and decayed by multiplying a factor of 0.9 after every one-tenth of the total epochs. The number of training epochs is set to 300.
6) *SSFTT [25]:* This model integrates both CNN and transformer architectures. It leverages 3-D and 2-D convolutions to extract shallow spectral and spatial features, while employing a transformer module to model high-level semantic representations. The learning rate is set to $1e^{-3}$, the batch size is set to 64, and the number of training epochs is set to 100.
7) *HiT [26]:* This model designs a spectral-adaptive 3-D convolution projection module for shallow feature mapping, which is then fed into transformer encoders with depthwise convolution and pointwise convolution to extract spatial-spectral features. This model adopts the Adam optimizer with a batch size of 100 and the learning rate $1e^{-3}$. The number of training epochs is set to 100.
8) *SS-TMNet [27]:* This model employs a multiscale 3-D convolution module to extract shallow spatial-spectral features. These features are then fed into transformer encoders equipped with height spatial attention, width spatial attention, and spectral attention mechanisms to capture global spatial-spectral features. This model uses the Adam optimizer for gradient descent and sets the initial learning rate to $1e^{-4}$. The mini-batch size is set to 32, and the number of training epochs is set to 200.
9) *MorphFormer [23]:* This model integrates morphology and transformer architectures, leveraging morphological convolutions to fuse HSI tokens and class tokens from both spatial and spectral perspectives. The batch size is set to 64. The Adam optimizer has been used, with a weight decay of $5e^{-3}$ and learning rate of $5e^{-4}$. It uses a step scheduler with a gamma of 0.9, steps of size 50, and is trained for 500 epochs.
10) *GSCViT [24]:* This model integrates the groupwise separable convolution module and groupwise separable multihead self-attention module, effectively reducing parameters while extracting local and global spectral-spatial information. AdamW optimizer is employed for parameter updating over a span of 200 training epochs. The learning rate is set to 0.001, and a weight decay coefficient of 0.05 is applied. The batch size is set to 128.
11) *S2VNet [30]:* This model integrates the classification and unmixing tasks into a unified framework. The unmixing branch can alleviate the mixed pixel and provide rich multilevel information to support accurate classification. The learning rate is set to $1e^{-3}$ and decays by multiplying a factor of 0.9 after each 50 epochs. The Adam optimizer is utilized to update the parameters with a minibatch size of 64. The number of training epochs is 500.
12) *MCTGCL [48]:* This model contains a lightweight dual-branch CNN-transformer network and adopts the graph contrastive learning to the topological structure of labeled samples. This model sets the total number of training epochs to 100. The learning rate is 0.001 and the batch size is 64.

C. Implementation Details

The proposed method is implemented using the PyTorch framework. In our experiment, we randomly select 10 samples per class for the training set, 5% of the remaining samples are used as the validation set, and the rest are allocated to the test set. The input patch size is set to $16 \times 16 \times B$, where B represents the number of spectral bands in the original HSIs. In the backbone network, the layer numbers in the transformer block are {2,2,6,2}, the layer numbers in the CNN block are {3,4,6,3}, and the channel dimension is set to 96. In task-specific branches, the layer number in the transformer is set to 2, and the channel dimension is set to 256. For the loss function, we set $\lambda_1 = 0.1$ and $\lambda_2 = 1e^{-4}$. For the training settings, we employ the AdamW optimizer to train the network for 200 epochs. The learning rate is set to $5e^{-5}$ and the size of the mini-batch is set to 128. All experiments are implemented on an NVIDIA RTX 4090 GPU. We perform the experiments

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TABLE I

QUANTITATIVE CLASSIFICATION RESULTS OF THE INDIAN PINES DATASET WITH TEN LABELED TRAINING SAMPLES PER CLASS. BEST RESULTS ARE SHOWN IN BOLD

Class	M3D-DCNN	3D-CNN	DFFN	RSSAN	SpectralFormer	SSFTT	HiT	SS-TMNet	MorphFormer	GSCViT	S2VNet	MCTGCL	Ours
1	72.94 ± 37.18	91.18 ± 6.96	94.41 ± 5.33	82.35 ± 11.69	94.71 ± 6.94	**100.00 ± 0.00**	95.88 ± 4.20	99.41 ± 1.76	99.71 ± 0.88	**100.00 ± 0.00**	97.94±3.49	**100.00±0.00**	99.76 ± 0.73
2	22.35 ± 12.96	19.45 ± 13.53	48.54 ± 10.74	25.40 ± 10.44	35.86 ± 6.75	63.70 ± 4.98	35.02 ± 9.65	40.28 ± 10.00	65.09 ± 3.70	63.10 ± 8.54	63.77±11.26	70.03±8.56	**70.25 ± 9.41**
3	24.90 ± 13.40	22.59 ± 13.32	45.28 ± 14.86	27.68 ± 8.28	42.73 ± 9.42	65.31 ± 8.22	33.59 ± 10.90	51.96 ± 8.59	68.96 ± 7.10	**75.30 ± 6.44**	77.27±9.73	71.62±5.41	74.98 ± 9.85
4	40.23 ± 19.37	33.61 ± 12.87	89.81 ± 6.79	38.66 ± 12.06	66.57 ± 10.00	95.93 ± 3.93	63.70 ± 11.25	94.54 ± 4.12	94.68 ± 6.59	**97.08 ± 6.33**	94.95±3.19	95.28±6.55	96.41 ± 4.81
5	28.55 ± 16.06	26.68 ± 11.02	68.49 ± 9.90	40.22 ± 16.10	41.20 ± 14.40	80.60 ± 11.47	39.80 ± 17.82	66.48 ± 9.07	74.21 ± 13.02	79.78 ± 9.40	73.47±13.11	73.01±7.71	**79.47 ± 9.19**
6	51.52 ± 29.80	69.47 ± 10.47	73.93 ± 10.98	34.37 ± 13.15	58.63 ± 9.95	93.76 ± 4.27	64.31 ± 6.74	87.46 ± 7.32	88.86 ± 5.24	94.12 ± 4.00	91.84±5.22	**95.03±2.64**	93.15 ± 2.43
7	72.35 ± 38.31	95.88 ± 3.77	**100.00 ± 0.00**	86.47 ± 16.85	**100.00 ± 0.00**	**100.00 ± 0.00**	98.82 ± 2.35	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00±0.00**	**100.00±0.00**	**100.00 ± 0.00**
8	71.66 ± 36.59	90.94 ± 3.20	93.69 ± 3.88	74.22 ± 20.30	92.58 ± 3.37	98.85 ± 1.79	91.06 ± 5.29	99.03 ± 1.07	96.70 ± 4.20	99.46 ± 0.89	97.28±3.19	**99.66±0.59**	99.28 ± 1.08
9	78.00 ± 39.45	94.00 ± 9.17	**100.00 ± 0.00**	91.00 ± 13.00	95.00 ± 12.04	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00±0.00**	**100.00±0.00**	**100.00 ± 0.00**
10	31.47 ± 19.17	18.42 ± 16.39	59.07 ± 13.33	33.18 ± 10.77	52.01 ± 12.90	70.37 ± 4.37	41.97 ± 8.59	61.64 ± 6.63	69.75 ± 5.93	77.59 ± 5.80	74.00±6.40	**83.81±4.99**	80.94 ± 6.48
11	47.10 ± 23.07	65.60 ± 13.86	59.25 ± 11.16	43.70 ± 17.47	56.11 ± 7.13	68.13 ± 6.28	52.06 ± 5.58	67.92 ± 11.15	65.91 ± 7.21	69.60 ± 8.71	74.82±9.38	73.41±9.85	**83.30 ± 4.99**
12	22.78 ± 13.10	23.59 ± 8.16	54.86 ± 10.10	29.57 ± 4.73	36.99 ± 7.30	64.80 ± 6.24	37.65 ± 10.81	55.29 ± 9.44	62.58 ± 7.24	67.04 ± 7.09	64.42±8.39	73.54±5.08	**77.21 ± 7.88**
13	70.32 ± 36.78	86.59 ± 14.14	95.73 ± 2.84	72.22 ± 18.12	91.03 ± 7.69	98.92 ± 1.79	92.49 ± 6.88	99.35 ± 0.76	**99.57 ± 0.76**	96.86 ± 6.27	98.49±2.17	97.95±2.63	98.80 ± 1.83
14	76.38 ± 27.24	80.43 ± 5.21	80.66 ± 8.80	64.60 ± 21.13	78.11 ± 7.11	**94.12 ± 3.42**	77.78 ± 9.59	86.48 ± 3.52	86.69 ± 6.02	92.59 ± 5.67	92.53±5.03	90.96±4.59	92.98 ± 5.24
15	44.20 ± 21.85	35.80 ± 12.05	73.47 ± 14.27	37.06 ± 18.53	69.50 ± 11.94	88.80 ± 7.42	57.90 ± 10.63	79.86 ± 5.78	86.75 ± 8.26	90.08 ± 7.38	**87.90±10.22**	81.48±5.30	85.79 ± 17.95
16	58.48 ± 32.89	82.15 ± 7.91	90.51 ± 11.52	63.54 ± 18.35	94.05 ± 4.94	96.33 ± 4.47	90.89 ± 8.60	**96.96 ± 4.21**	96.58 ± 2.72	96.20 ± 2.59	98.10±1.30	95.95±5.18	95.24 ± 3.88
OA(%)	43.60 ± 15.94	49.05 ± 2.98	64.50 ± 3.90	41.81 ± 7.89	56.57 ± 1.36	76.80 ± 1.84	53.55 ± 2.01	68.58 ± 2.99	74.80 ± 2.92	78.56 ± 2.88	78.93±3.03	81.43±2.91	**83.82 ± 2.17**
AA(%)	50.83 ± 21.93	58.53 ± 3.05	76.73 ± 3.29	52.77 ± 7.91	69.07 ± 1.59	86.23 ± 1.32	67.06 ± 2.62	80.42 ± 1.85	84.75 ± 1.90	87.43 ± 1.42	86.67±1.53	88.82±1.13	**89.22 ± 1.87**
Kappa	0.37 ± 0.17	0.41 ± 0.04	0.60 ± 0.05	0.35 ± 0.08	0.51 ± 0.01	0.74 ± 0.02	0.48 ± 0.02	0.65 ± 0.03	0.72 ± 0.03	0.76 ± 0.03	0.76±0.03	0.79±0.03	**0.82 ± 0.02**

TABLE II

QUANTITATIVE CLASSIFICATION RESULTS OF THE SALINAS DATASET WITH TEN LABELED TRAINING SAMPLES PER CLASS. BEST RESULTS ARE SHOWN IN BOLD

Class	M3D-DCNN	3D-CNN	DFFN	RSSAN	SpectralFormer	SSFTT	HiT	SS-TMNet	MorphFormer	GSCViT	S2VNet	MCTGCL	Ours
1	83.67 ± 28.82	80.96 ± 19.92	91.42 ± 11.68	84.80 ± 17.61	77.04 ± 11.63	95.13 ± 6.39	88.56 ± 8.92	99.67 ± 0.58	99.59 ± 0.61	98.74 ± 2.52	99.67 ± 0.63	99.59 ± 0.80	**99.99 ± 0.02**
2	81.86 ± 25.63	79.56 ± 13.30	96.02 ± 3.03	82.24 ± 16.34	89.38 ± 6.22	86.36 ± 22.12	85.44 ± 11.32	97.38 ± 3.51	98.75 ± 1.74	98.99 ± 1.28	99.55 ± 0.62	**99.97 ± 0.09**	99.93 ± 0.15
3	80.89 ± 29.21	65.42 ± 28.77	92.43 ± 5.93	88.07 ± 6.22	87.14 ± 9.01	88.39 ± 12.29	90.75 ± 8.17	95.86 ± 4.22	95.73 ± 3.76	96.40 ± 4.18	98.06 ± 2.99	97.99 ± 0.02	**98.87 ± 1.90**
4	94.93 ± 7.00	89.42 ± 9.48	96.21 ± 3.84	93.71 ± 4.35	96.11 ± 2.47	94.97 ± 7.92	97.16 ± 2.07	98.05 ± 3.43	97.88 ± 2.49	97.48 ± 4.29	99.19 ± 1.14	98.55 ± 1.44	**99.45 ± 0.85**
5	90.26 ± 11.05	79.87 ± 14.37	88.70 ± 7.05	92.06 ± 11.11	88.85 ± 10.50	89.57 ± 9.43	95.51 ± 3.13	96.02 ± 5.05	92.07 ± 7.27	96.33 ± 3.04	96.15 ± 4.22	**97.15 ± 3.62**	96.35 ± 3.00
6	98.97 ± 1.78	98.01 ± 2.79	97.13 ± 4.70	95.82 ± 4.27	98.43 ± 2.14	97.25 ± 3.89	98.85 ± 1.93	99.31 ± 1.20	99.11 ± 1.29	97.71 ± 3.19	**99.91 ± 0.17**	99.49 ± 0.66	99.67 ± 0.43
7	92.32 ± 3.68	69.95 ± 28.83	93.83 ± 5.96	87.78 ± 6.97	93.19 ± 4.14	97.51 ± 2.38	93.84 ± 3.26	97.70 ± 2.39	98.78 ± 1.57	98.74 ± 1.92	99.70 ± 0.43	**99.95 ± 0.10**	99.93 ± 0.14
8	74.18 ± 10.48	79.11 ± 9.86	75.73 ± 7.71	59.81 ± 11.56	73.28 ± 4.45	65.57 ± 19.15	68.36 ± 8.45	74.38 ± 3.97	81.51 ± 5.41	77.22 ± 8.10	82.19 ± 4.55	82.00 ± 4.64	**84.37 ± 8.57**
9	92.46 ± 14.15	93.50 ± 8.26	98.63 ± 1.52	93.84 ± 2.91	97.02 ± 2.48	95.45 ± 2.34	98.34 ± 2.76	99.66 ± 0.53	98.99 ± 1.00	99.41 ± 0.73	99.97 ± 0.05	**100.00 ± 0.00**	99.76 ± 0.51
10	83.25 ± 11.23	75.68 ± 19.40	87.27 ± 8.06	79.56 ± 14.09	87.85 ± 6.07	81.19 ± 13.81	86.76 ± 8.73	94.82 ± 4.34	94.47 ± 3.08	94.81 ± 4.59	93.12 ± 8.15	**96.83 ± 2.03**	95.50 ± 4.24
11	80.00 ± 15.91	70.24 ± 14.86	92.24 ± 18.19	84.59 ± 14.41	81.92 ± 13.90	91.99 ± 7.75	93.35 ± 8.83	99.49 ± 0.72	99.27 ± 1.04	98.13 ± 4.45	97.63 ± 3.47	99.90 ± 0.17	**99.99 ± 0.03**
12	97.63 ± 3.33	86.47 ± 13.92	99.18 ± 1.07	95.23 ± 3.31	93.79 ± 5.34	90.15 ± 14.06	98.60 ± 0.93	99.24 ± 1.64	99.12 ± 0.92	81.67 ± 18.73	99.34 ± 1.21	98.97 ± 1.07	**99.59 ± 0.76**
13	96.81 ± 3.66	96.98 ± 2.69	97.50 ± 1.89	92.40 ± 8.62	97.34 ± 3.02	94.23 ± 9.78	99.07 ± 1.02	99.87 ± 0.21	98.46 ± 1.46	83.89 ± 18.41	99.71 ± 0.69	96.85 ± 5.92	**99.52 ± 1.15**
14	93.91 ± 4.20	92.02 ± 4.38	99.67 ± 0.79	94.98 ± 2.70	98.20 ± 1.67	91.50 ± 9.31	99.28 ± 0.76	99.78 ± 0.31	98.50 ± 1.72	95.94 ± 9.94	99.73 ± 0.46	98.57 ± 1.50	**99.88 ± 0.19**
15	49.87 ± 24.71	31.63 ± 22.90	83.52 ± 8.36	44.59 ± 15.14	79.75 ± 5.24	63.81 ± 24.98	72.26 ± 13.10	75.17 ± 10.38	85.83 ± 9.45	84.87 ± 11.69	84.19 ± 7.75	84.04 ± 4.66	**91.66 ± 5.82**
16	84.98 ± 8.42	75.63 ± 9.82	89.09 ± 4.44	93.40 ± 4.25	91.96 ± 5.42	94.45 ± 6.74	89.17 ± 7.14	98.26 ± 3.99	98.44 ± 1.54	98.97 ± 2.06	99.26 ± 1.26	**99.90 ± 0.12**	99.66 ± 0.44
OA(%)	81.10 ± 7.78	75.37 ± 4.25	89.06 ± 1.51	77.41 ± 1.96	86.43 ± 1.50	82.78 ± 2.82	85.56 ± 2.27	90.07 ± 1.45	92.78 ± 1.14	91.01 ± 1.88	93.28 ± 1.32	93.03 ± 0.89	**94.99 ± 1.71**
AA(%)	86.00 ± 8.85	79.03 ± 5.78	92.41 ± 2.11	85.18 ± 2.46	89.45 ± 1.30	88.60 ± 2.58	90.96 ± 1.87	95.29 ± 0.93	96.03 ± 0.71	93.71 ± 1.37	96.71 ± 0.58	97.67 ± 0.46	**97.76 ± 0.57**
Kappa	0.79 ± 0.09	0.72 ± 0.05	0.88 ± 0.02	0.75 ± 0.02	0.85 ± 0.02	0.81 ± 0.03	0.84 ± 0.03	0.89 ± 0.02	0.92 ± 0.01	0.90 ± 0.02	0.93 ± 0.01	0.94 ± 0.01	**0.94 ± 0.02**

ten times and present the results as mean ± standard deviation to avoid disturbance caused by initialization.

D. Classification Results

In this section, we conduct experiments on four datasets: Indian Pines, Salinas, Kennedy Space Center, and WHU-Hi-HanChuan. To further demonstrate the effectiveness and generalization capability of our method, we provide both quantitative analysis and visual evaluation.

1) Quantitative Analysis: Tables I–IV summarize the OA, AA, Kappa, and the accuracy of each class for all compared methods on four HSI datasets. The best results are highlighted in bold. It can be clearly observed that our method achieves the best results across four datasets.

In the Indian Pines dataset, the OA metric reached 83.83% ± 2.17% using only ten training samples per class, which is about 5% higher than the second place. Although our method may not outperform some SOTA methods in terms of classification accuracy for some specific classes, it achieves the highest OA, indicating superior generalization performance. It is worth noting that CNN-based methods [18], [19], [46], [47] generally exhibited inferior performance compared to transformer-based methods [22], [23], [24]. This is primarily due to the limited receptive field of convolutional kernels, which constrains them to capturing only local information and hinders their ability to model global contextual dependencies. Our method integrates the strengths of both CNN and transformer architectures. It can be found that the CNN-transformer framework can effectively improve classification accuracy to some extent, like SSFTT [25]. However, simply merging the two networks does not fully leverage their individual strengths and may even lead to performance degradation, as observed in HiT [26] and SS-TMNet [27] in Table I. Different from existing hybrid CNN-transformer frameworks, we introduce an FIM to enhance the fusion of local and global features from a frequency perspective. Experimental results further validate the effectiveness of our proposed approach.

For the Salinas, Kennedy Space Center, and WHU-Hi-HanChuan datasets, we achieved OA of 94.99% ± 1.71%, 91.68% ± 2.65%, and 87.49% ± 1.94%, outperforming the methods in the second place by about 2%, 3%, and 4%, respectively. It can be observed that the performance on the three datasets is consistently higher than that on the Indian Pines dataset. In our experimental setup, we ensure that the sample size for each class is consistent across all four datasets, reducing the impact of class imbalance. Therefore, we attribute the performance discrepancy primarily to the differences in spatial resolution between the datasets. The Indian Pines

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TABLE III

QUANTITATIVE CLASSIFICATION RESULTS OF THE KENNEDY SPACE CENTER DATASET WITH TEN LABELED TRAINING SAMPLES PER CLASS. BEST RESULTS ARE SHOWN IN BOLD

Class	M3D-DCNN	3D-CNN	DFFN	RSSAN	SpectralFormer	SSFTT	HiT	SS-TMNet	MorphFormer	GSCViT	S2VNet	MCTGCL	Ours
1	86.40 ± 11.06	75.52 ± 38.02	94.71 ± 7.07	68.56 ± 7.01	92.56 ± 3.21	92.89 ± 8.32	85.92 ± 6.82	85.34 ± 15.85	97.14 ± 1.91	95.35 ± 6.49	94.08±3.70	87.86±8.51	**97.72 ± 2.08**
2	36.79 ± 12.34	33.03 ± 8.47	67.29 ± 13.68	13.26 ± 6.41	19.37 ± 10.00	22.53 ± 21.02	24.75 ± 7.87	52.85 ± 9.27	40.81 ± 16.42	75.40 ± 8.48	76.83±11.96	74.36±9.41	**79.19 ± 16.99**
3	33.21 ± 28.26	17.86 ± 30.05	78.33 ± 16.86	37.74 ± 8.82	44.36 ± 26.01	12.95 ± 29.02	52.95 ± 18.67	66.71 ± 21.59	20.98 ± 22.47	80.00 ± 19.94	76.88±10.42	74.36±29.41	**86.75 ± 11.43**
4	19.39 ± 13.29	6.87 ± 5.10	59.39 ± 13.94	19.26 ± 7.24	42.48 ± 13.26	20.30 ± 6.93	21.78 ± 8.61	35.17 ± 17.95	44.74 ± 16.60	56.47 ± 16.73	40.30±12.32	37.83±10.60	**63.91 ± 18.47**
5	61.54 ± 10.50	45.24 ± 14.39	74.48 ± 14.53	48.74 ± 14.45	52.38 ± 4.92	52.03 ± 11.11	68.74 ± 13.01	74.90 ± 10.90	59.09 ± 12.03	74.19 ± 16.21	73.43±12.31	65.03±8.21	**85.38 ± 8.73**
6	25.48 ± 16.98	13.80 ± 4.23	82.16 ± 12.55	44.57 ± 14.75	21.54 ± 19.71	23.61 ± 25.66	56.44 ± 11.91	55.29 ± 22.48	29.33 ± 21.58	76.62 ± 15.64	69.66±15.19	61.11±3.71	**85.77 ± 8.73**
7	47.00 ± 25.47	48.78 ± 32.96	97.89 ± 3.04	73.00 ± 17.38	31.67 ± 31.23	21.56 ± 34.72	81.00 ± 17.56	98.33 ± 3.92	28.22 ± 31.09	97.89 ± 2.98	98.56±2.98	92.22±2.47	**99.22 ± 1.41**
8	40.83 ± 29.00	14.72 ± 16.10	77.25 ± 14.59	39.23 ± 13.76	61.23 ± 14.56	42.42 ± 34.68	50.40 ± 9.82	83.25 ± 17.22	70.28 ± 16.83	82.37 ± 9.12	69.95±6.01	71.15±3.90	**83.53 ± 16.42**
9	62.52 ± 38.21	73.13 ± 37.10	89.07 ± 7.89	**93.36 ± 4.34**	65.98 ± 11.02	36.02 ± 37.60	86.91 ± 4.50	88.74 ± 5.76	86.74 ± 8.97	86.01 ± 7.83	93.59±5.52	53.03±22.41	89.63 ± 9.51
10	36.71 ± 12.49	23.26 ± 12.46	85.67 ± 11.30	9.04 ± 3.74	64.92 ± 10.91	72.51 ± 21.50	38.50 ± 10.73	79.47 ± 9.56	80.03 ± 9.77	93.35 ± 8.56	69.06±17.33	73.80±8.67	**98.66 ± 1.34**
11	54.24 ± 31.65	60.23 ± 27.92	93.32 ± 7.73	66.86 ± 12.92	79.05 ± 16.67	90.69 ± 11.48	95.91 ± 3.11	96.38 ± 7.03	94.68 ± 3.88	97.10 ± 3.94	96.53±3.74	86.30±9.08	**99.64 ± 0.57**
12	41.54 ± 11.20	34.06 ± 9.57	78.55 ± 7.23	15.24 ± 9.99	76.56 ± 10.09	60.58 ± 21.93	39.42 ± 11.99	86.15 ± 9.38	75.28 ± 15.87	85.88 ± 10.73	85.43±11.73	79.94±7.47	**93.89 ± 5.76**
13	43.27 ± 30.33	61.48 ± 29.14	96.96 ± 3.10	93.97 ± 5.07	**99.86 ± 0.41**	98.74 ± 2.35	76.44 ± 11.83	95.52 ± 8.44	99.79 ± 0.39	99.09 ± 1.88	99.62±0.63	94.39±4.86	97.39 ± 2.53
OA(%)	49.51 ± 13.07	46.34 ± 11.93	85.91 ± 3.13	55.37 ± 3.54	70.37 ± 3.39	63.08 ± 5.51	64.66 ± 3.55	81.72 ± 4.75	76.60 ± 3.12	88.13 ± 2.14	84.68±1.30	78.86±7.44	**91.68 ± 2.65**
AA(%)	45.30 ± 13.06	39.08 ± 10.48	82.70 ± 3.45	47.91 ± 4.31	57.84 ± 5.10	49.76 ± 7.68	59.94 ± 3.97	76.78 ± 4.48	63.62 ± 6.65	84.59 ± 2.43	80.30±1.79	69.75±6.84	**89.28 ± 3.07**
Kappa	0.44 ± 0.15	0.40 ± 0.13	0.84 ± 0.03	0.50 ± 0.04	0.67 ± 0.04	0.58 ± 0.06	0.61 ± 0.04	0.80 ± 0.05	0.74 ± 0.04	0.87 ± 0.02	0.83±0.01	0.75±0.08	**0.91 ± 0.03**

TABLE IV

QUANTITATIVE CLASSIFICATION RESULTS OF THE WHU-HI-HANCHUAN DATASET WITH TEN LABELED TRAINING SAMPLES PER CLASS. BEST RESULTS ARE SHOWN IN BOLD

Class	M3D-DCNN	3D-CNN	DFFN	RSSAN	SpectralFormer	SSFTT	HiT	SS-TMNet	MorphFormer	GSCViT	S2VNet	MCTGCL	Ours
1	55.13 ± 22.08	56.48 ± 8.37	75.41 ± 7.89	63.77 ± 15.51	80.24 ± 5.60	78.98 ± 9.99	67.68 ± 11.45	81.74 ± 6.38	86.04 ± 6.81	86.21 ± 6.53	89.62±7.41	87.70±7.32	**89.56 ± 6.94**
2	52.48 ± 12.30	56.11 ± 9.44	66.67 ± 6.82	50.16 ± 12.46	55.17 ± 4.63	73.14 ± 13.51	69.42 ± 5.47	80.34 ± 6.12	77.98 ± 8.23	68.32 ± 14.28	58.77±4.74	81.53±4.22	**82.57 ± 4.51**
3	20.69 ± 9.92	38.61 ± 11.31	44.15 ± 14.50	33.98 ± 11.12	62.19 ± 9.88	59.31 ± 13.40	36.84 ± 15.02	62.62 ± 12.17	75.76 ± 6.94	76.67 ± 8.99	74.72±6.23	**88.92±4.65**	82.20 ± 13.05
4	78.11 ± 20.77	50.85 ± 25.18	84.16 ± 7.84	73.24 ± 10.21	78.14 ± 9.18	91.38 ± 4.69	77.38 ± 11.30	94.02 ± 2.33	95.35 ± 4.68	88.23 ± 15.52	89.29±14.04	**97.65±1.41**	97.04 ± 1.35
5	54.74 ± 24.88	77.11 ± 17.29	84.82 ± 22.54	63.39 ± 13.56	88.82 ± 7.03	93.08 ± 5.85	82.72 ± 19.65	93.75 ± 6.62	96.73 ± 2.95	83.37 ± 27.55	97.44±2.26	97.29±2.05	**97.81 ± 2.37**
6	19.99 ± 8.36	16.03 ± 8.76	36.29 ± 14.17	16.12 ± 4.14	40.35 ± 7.48	37.37 ± 12.16	23.27 ± 8.12	40.79 ± 9.06	56.46 ± 10.47	46.70 ± 18.96	40.24±10.76	57.77±3.67	**65.35 ± 17.46**
7	76.86 ± 13.44	86.98 ± 7.08	68.17 ± 21.93	71.57 ± 9.79	73.69 ± 13.51	80.67 ± 9.92	85.51 ± 9.08	83.54 ± 25.56	88.25 ± 5.45	79.47 ± 12.92	80.62±10.61	91.45±3.98	**93.03 ± 4.54**
8	30.94 ± 9.11	33.32 ± 11.75	51.24 ± 8.29	21.86 ± 10.61	49.47 ± 8.87	63.03 ± 14.08	43.61 ± 10.17	51.41 ± 7.92	56.98 ± 8.04	57.88 ± 10.86	51.72±9.44	**70.82±6.41**	70.59 ± 6.99
9	25.38 ± 10.19	27.14 ± 5.05	40.93 ± 12.32	26.92 ± 9.41	39.30 ± 6.70	64.50 ± 17.69	42.86 ± 14.83	52.22 ± 11.96	65.76 ± 10.20	63.00 ± 17.05	54.76±10.31	73.30±11.93	**74.94 ± 11.15**
10	51.34 ± 11.51	53.12 ± 16.52	63.46 ± 15.59	52.21 ± 13.13	77.65 ± 12.24	73.66 ± 17.54	64.07 ± 23.43	76.33 ± 17.61	85.12 ± 10.20	84.30 ± 12.98	91.13±4.76	89.54±2.13	**90.99 ± 6.75**
11	69.67 ± 16.78	77.30 ± 13.27	64.60 ± 14.41	48.00 ± 21.58	73.65 ± 8.20	60.43 ± 24.78	70.97 ± 13.18	84.12 ± 13.89	81.12 ± 16.08	71.19 ± 16.18	76.95±13.13	88.93±9.32	**89.73 ± 6.92**
12	33.43 ± 9.67	48.84 ± 14.97	69.08 ± 11.75	42.35 ± 17.33	64.84 ± 11.69	77.39 ± 13.32	49.55 ± 12.93	62.12 ± 15.33	85.56 ± 10.83	79.10 ± 18.24	66.87±14.13	**89.56±8.90**	85.53 ± 11.29
13	43.11 ± 11.34	40.23 ± 9.31	48.11 ± 8.22	34.75 ± 8.61	49.29 ± 6.73	43.32 ± 14.72	48.29 ± 8.26	50.25 ± 19.38	64.71 ± 9.84	58.26 ± 10.46	57.02±6.16	65.06±8.95	**70.98 ± 9.25**
14	58.34 ± 13.48	62.58 ± 11.31	66.06 ± 12.59	54.51 ± 9.35	62.49 ± 10.17	61.55 ± 15.27	72.52 ± 12.58	66.21 ± 18.71	72.57 ± 10.01	71.86 ± 10.97	70.66±10.38	72.05±8.32	**85.20 ± 5.27**
15	60.32 ± 14.50	49.91 ± 9.99	77.65 ± 6.29	54.50 ± 8.39	76.35 ± 6.35	**90.28 ± 7.62**	80.54 ± 7.52	89.16 ± 4.45	88.85 ± 5.38	85.16 ± 8.09	78.85±8.09	87.42±3.10	87.07 ± 5.21
16	88.84 ± 9.91	74.68 ± 5.03	94.63 ± 4.00	92.23 ± 4.83	90.89 ± 7.21	93.11 ± 7.87	91.24 ± 7.16	91.61 ± 7.22	**98.18 ± 1.31**	95.14 ± 6.33	94.99±7.59	95.94±3.13	96.12 ± 4.71
OA(%)	61.29 ± 5.79	59.29 ± 1.18	72.52 ± 2.32	61.43 ± 3.56	72.77 ± 2.28	75.88 ± 2.17	70.34 ± 5.43	77.63 ± 6.02	83.39 ± 2.13	80.02 ± 2.93	79.24±2.53	85.41±2.36	**87.49 ± 1.94**
AA(%)	51.21 ± 7.22	53.08 ± 3.99	64.72 ± 3.02	49.97 ± 3.91	66.41 ± 2.62	71.32 ± 2.46	62.90 ± 6.51	72.51 ± 8.48	79.71 ± 2.48	74.68 ± 4.88	73.35±2.62	84.25±1.35	**84.92 ± 2.04**
Kappa	0.56 ± 0.06	0.54 ± 0.01	0.68 ± 0.03	0.55 ± 0.04	0.69 ± 0.03	0.72 ± 0.02	0.66 ± 0.06	0.74 ± 0.07	0.81 ± 0.02	0.77 ± 0.03	0.76±0.03	0.83±0.03	**0.85 ± 0.02**

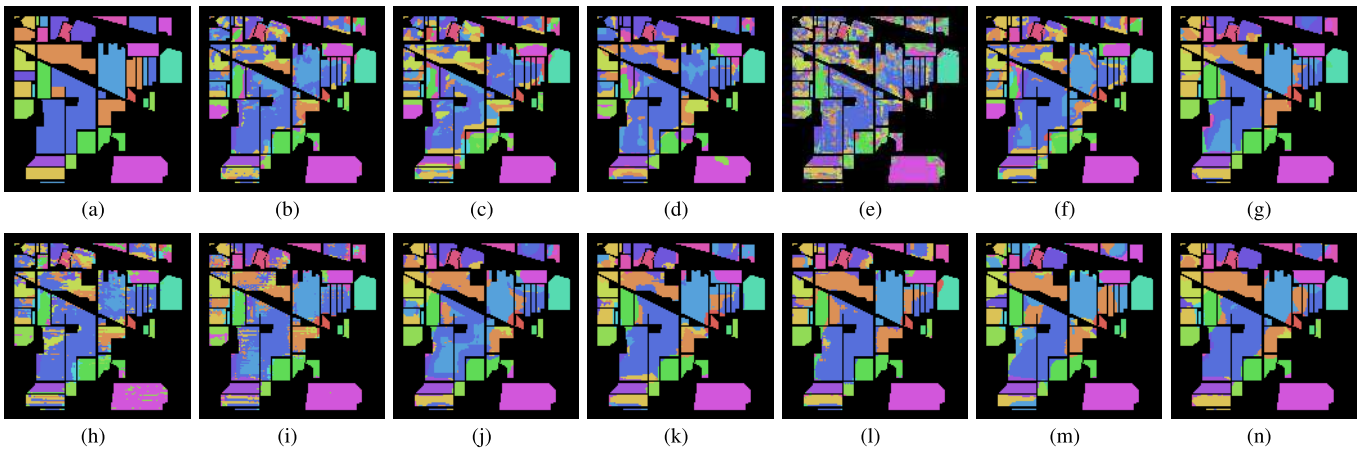


Fig. 6. Classification maps for the Indian Pines dataset. (a) GT. (b) M3DDCNN. (c) 3DCNN. (d) DFFN. (e) RSSAN. (f) SpectralFormer. (g) SSFTT. (h) HiT. (i) SSTMNet. (j) MorphFormer. (k) GSCViT. (l) S2VNet. (m) MCTGCL. (n) Ours.

dataset has a smaller spatial resolution and size compared to the other three datasets, leading to more severe mixed pixels and insufficient spatial information, which results in lower performance across all classifiers. Our proposed method, which incorporates the SR technique, helps to enhance spatial details to some extent, achieving the best classification performance.

2) Visual Evaluation: To further demonstrate the superiority of our proposed method, we conducted a qualitative analysis by visualizing the classification maps generated by different methods, as shown in Figs. 6–9. It can be clearly observed that the classification maps produced by our method contain less noise and are more consistent with the ground truth compared to those generated by other methods. This is mainly due to insufficient feature extraction, resulting in weak discriminative representations. The methods based solely on CNN or transformer tend to overlook either local or global information to varying degrees, thereby engendering noise. Our method integrates the strengths of both frameworks and

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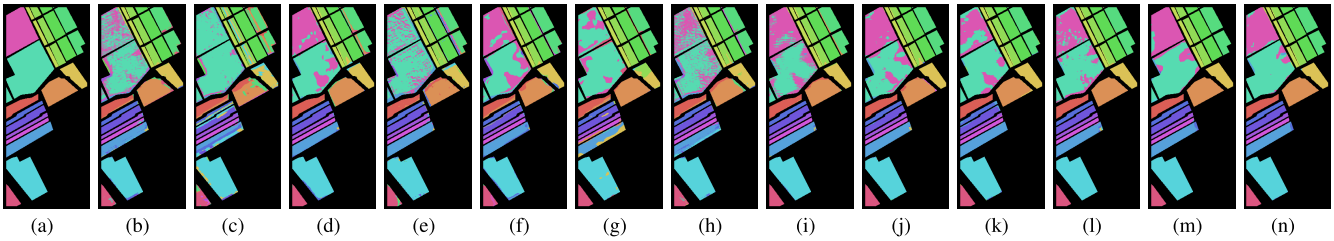


Fig. 7. Classification maps for the Salinas dataset. (a) GT. (b) M3DDCNN. (c) 3DCNN. (d) DFFN. (e) RSSAN. (f) SpectralFormer. (g) SSFTT. (h) HiT. (i) SSTMNet. (j) MorphFormer. (k) GSCViT. (l) S2VNet. (m) MCTGCL. (n) Ours.

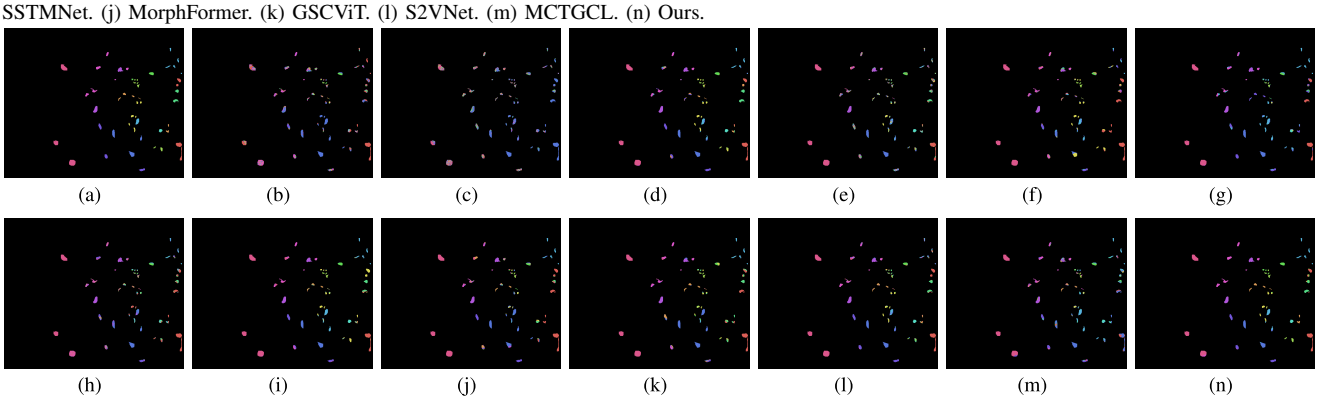


Fig. 8. Classification maps for the Kennedy Space Center dataset. (a) GT. (b) M3DDCNN. (c) 3DCNN. (d) DFFN. (e) RSSAN. (f) SpectralFormer. (g) SSFTT. (h) HiT. (i) SSTMNet. (j) MorphFormer. (k) GSCViT. (l) S2VNet. (m) MCTGCL. (n) Ours.

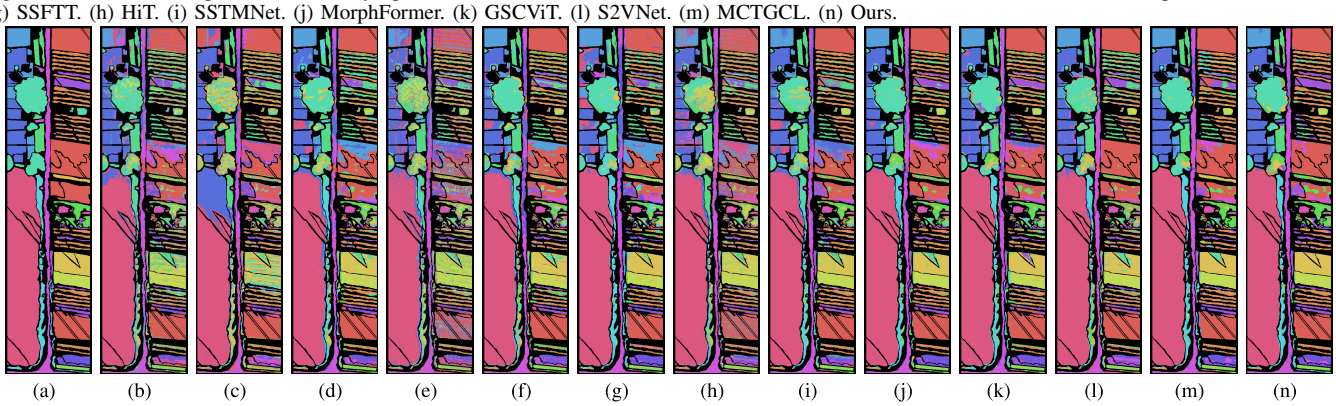


Fig. 9. Classification maps for the WHU-Hi-HanChuan dataset. (a) GT. (b) M3DDCNN. (c) 3DCNN. (d) DFFN. (e) RSSAN. (f) SpectralFormer. (g) SSFTT. (h) HiT. (i) SSTMNet. (j) MorphFormer. (k) GSCViT. (l) S2VNet. (m) MCTGCL. (n) Ours.

establishes information interaction from a frequency-domain perspective, effectively exploiting the redundant information inherent in HSIs to achieve better visual results. Notably, for most methods, misclassifications are mainly concentrated in boundary regions, which is particularly evident in the yellow triangular area of the WHU-Hi-HanChuan dataset. This may be because the information around class boundaries is often blurry, making it challenging for most methods to make accurate predictions. Our method integrates the SR task to enhance image quality and design a CTIM that enables the SR branch to provide more spatial details for the classification task, thereby achieving more accurate boundary predictions. This further demonstrates the effectiveness of the proposed framework.

E. Ablation Studies

In this section, we conduct a series of ablation experiments to verify the effectiveness of various modules through comparison between variants. All variants are trained ten times on four datasets. The results of the ablation experiments are presented in Table V. In the table, "SR" refers to the SR branch, "FIM" refers to the frequency-domain interaction module, "CTIM" refers to the cross-task interaction module, and "FCL" refers to the frequency-domain contrastive learning.

1) Ablation Study of the SR Branch: In our method, we introduce the SR branch to sharpen image edges and enrich spatial information, thereby enhancing classification performance. To verify the efficiency of the SR branch, we conduct experiments to quantitatively analyze the classification performance without and with the SR branch, as shown in the first and second rows of the table. It can be observed that the introduction of the SR branch leads to an increase in the OA metric across all four datasets. This finding aligns with the results in [7], where the performance improvement is achieved using SR features.

In addition, we further investigate an important parameter in the SR branch, namely the scale factor. We conduct experiments to evaluate the impact of different scale factors on

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TABLE V

COMPARISONS ON MODEL EFFICIENCY OF DIFFERENT METHODS ON FOUR DATASETS. BEST RESULTS ARE SHOWN IN BOLD

Dataset	SR	FIM	CTIM	FCL	OA(%)	AA(%)	Kappa
Indian Pines	-	-	-	-	80.81 ± 1.04	88.65 ± 0.53	0.79 ± 0.01
	✓	-	-	-	81.92 ± 1.55	87.74 ± 0.92	0.80 ± 0.02
	✓	✓	-	-	82.90 ± 1.26	89.39 ± 0.91	0.81 ± 0.01
	✓	✓	✓	-	83.45 ± 1.93	**89.42 ± 1.34**	0.81 ± 0.02
	✓	✓	✓	✓	**83.82 ± 2.17**	89.22 ± 1.87	**0.82 ± 0.02**
Salinas	-	-	-	-	92.46 ± 1.56	94.89 ± 0.56	0.91 ± 0.02
	✓	-	-	-	92.67 ± 1.01	94.91 ± 0.49	0.92 ± 0.01
	✓	✓	-	-	93.45 ± 1.91	96.62 ± 0.54	0.92 ± 0.02
	✓	✓	✓	-	94.21 ± 1.43	97.04 ± 0.46	0.93 ± 0.02
	✓	✓	✓	✓	**94.99 ± 1.71**	**97.76 ± 0.57**	**0.94 ± 0.02**
Kennedy Space Center	-	-	-	-	89.40 ± 1.35	87.23 ± 1.61	0.88 ± 0.02
	✓	-	-	-	90.04 ± 1.11	89.08 ± 1.35	0.89 ± 0.01
	✓	✓	-	-	90.51 ± 2.05	88.68 ± 2.57	0.90 ± 0.02
	✓	✓	✓	-	91.24 ± 1.75	88.91 ± 2.58	0.90 ± 0.02
	✓	✓	✓	✓	**91.68 ± 2.65**	**89.28 ± 3.07**	**0.91 ± 0.03**
WHU-Hi-HanChuan	-	-	-	-	85.60 ± 2.25	83.19 ± 2.14	0.82 ± 0.03
	✓	-	-	-	86.14 ± 2.78	84.19 ± 1.43	0.84 ± 0.03
	✓	✓	-	-	86.94 ± 1.90	84.39 ± 1.35	0.84 ± 0.02
	✓	✓	✓	-	87.13 ± 2.03	84.68 ± 1.93	0.84 ± 0.02
	✓	✓	✓	✓	**87.49 ± 1.94**	**84.92 ± 2.04**	**0.85 ± 0.02**

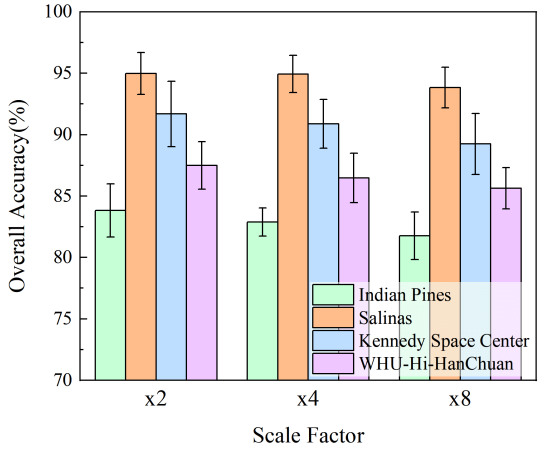


Fig. 10. Classification results of our method with different scale factors on four datasets.

classification performance across four datasets. As shown in Fig. 10, the classification performance shows little difference under different scale factors, with a slight decline as the scale factor increases. We argue that the benefit of SR for classification stems from the task itself, where most scale factors can enhance classification performance to varying degrees. However, excessively large scale factors may mislead the classification model. This is because SR tasks often struggle to maintain performance under large-scale factors, often introducing artifacts and other negative information, which may interfere with the decision-making process of the classifier. In addition, larger scale factors significantly increase GPU memory consumption. Therefore, after comprehensive consideration, the scale factor is set to 2.

2) Ablation Study of the Training Strategy: In our method, we adopt an end-to-end training strategy, which differs from the two-stage approach used in [7]. To verify the advantages of the end-to-end framework, we compare several two-stage frameworks, as illustrated in Fig. 11. They mainly contain two categories: one enriches image details through the SR method followed by the classification network, referred to as SR-Image; the other captures more discriminative features through the SR branch, referred to as SR-Feature. For the latter, considering that the frozen backbone (i.e., SR-Feature mode1) may hinder the feature extraction for the classification task, we introduce an additional trainable backbone module with the same architecture as SR-Feature mode2. As shown in Table VI, our end-to-end training strategy achieves competitive results. The serially concatenated SR-Image approach overlooks the connection between low-level and high-level visions, while the SR-Feature mode1 approach fully retains the SR structure but lacks adaptability to the classification task. It is worth noting that the SR-Feature mode2 method achieves performance comparable to ours. However, due to the introduction of an additional backbone network it results in increased parameters. Therefore, our end-to-end framework is both reasonable and effective.

TABLE VI

COMPARISONS ON MODEL EFFICIENCY OF DIFFERENT TRAINING STRATEGIES ON FOUR DATASETS. BEST RESULTS ARE SHOWN IN BOLD

		Two Stage			End-to-End
		SR-Image	SR-Feature mode1	SR-Feature mode2	Ours
Indian Pines	OA(%)	80.34 ±2.03	54.67 ±0.76	82.89 ±1.86	**83.82 ± 2.17**
	AA(%)	86.29 ±2.33	51.98 ±1.34	89.12 ±1.38	**89.22 ± 1.87**
	Kappa	0.78 ±0.02	0.56 ±0.02	0.81 ±0.02	**0.82 ± 0.02**
Salinas	OA(%)	92.49 ±1.55	83.51 ±1.18	94.53 ±1.66	**94.99 ± 1.71**
	AA(%)	95.97 ±0.66	87.76 ±1.82	97.55 ±0.52	**97.76 ± 0.57**
	Kappa	0.92 ±0.02	0.82 ±0.01	0.93 ±0.02	**0.94 ± 0.02**
Kennedy Space Center	OA(%)	70.96 ±3.37	58.21 ±2.33	**91.84 ±1.50**	91.68 ± 2.65
	AA(%)	62.02 ±4.52	44.00 ±1.85	**89.29 ±2.03**	89.28 ± 3.07
	Kappa	0.68 ±0.04	0.53 ±0.02	**0.92 ±0.02**	0.91 ± 0.03
WHU-Hi-HanChuan	OA(%)	85.44 ±1.95	56.14 ±1.89	87.42 ±1.95	**87.49 ± 1.94**
	AA(%)	82.31 ±1.60	54.74 ±1.80	84.78 ±1.56	**84.92 ± 2.04**
	Kappa	0.83 ±0.02	0.52 ±0.02	0.84 ±0.02	**0.85 ± 0.02**

3) Ablation Study of the FIM: In our method, we exploit the strengths of both CNN and transformer architectures. According to the analysis in Section IV-D, simply concatenating two networks fails to fully leverage their respective strengths and may even lead to performance degradation. To alleviate this issue, we exploit the differences between CNN and transformer in frequency-domain feature extraction by designing an FIM to achieve more powerful representation capabilities. As observed from Table V, the classification performance is further improved with the incorporation of the FIM, demonstrating the effectiveness of the proposed module.

4) Ablation Study of the CTIM: Simply integrating the SR task and the classification task into a unified framework can enrich the diversity of feature extraction, but it neglects the beneficial impact of the SR task on classification performance. Considering this, we designed a CTIM to strengthen the correlation between tasks. As shown in Table V, it can be found that the OA metric is further improved, which demonstrates the significance of the proposed module.

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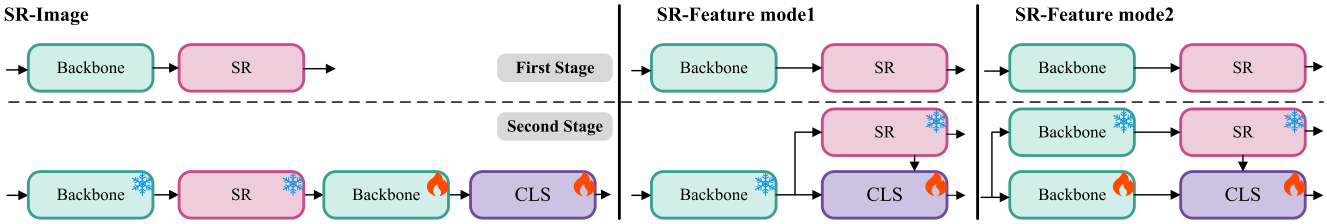


Fig. 11. Different two-stage training approaches. Three different variants are presented, including SR-Image, SR-Feature mode1, and SR-Feature mode2. The snowflake denotes the module with pre-trained weights that are frozen, while the spark indicates the randomly initialized trainable module.

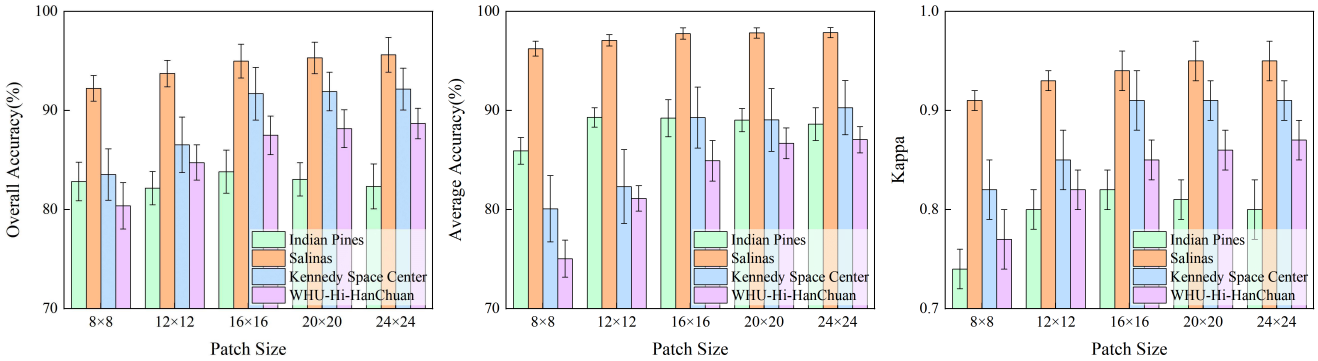


Fig. 12. OA, AA, and Kappa of different patch sizes on four datasets.

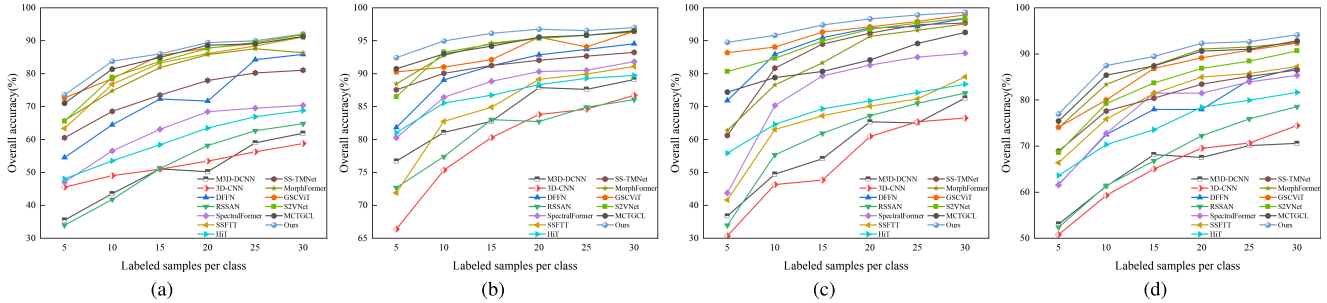


Fig. 13. OA curves of different numbers of labeled samples. (a) Indian Pines. (b) Salinas. (c) Kennedy Space Center. (d) WHU-Hi-HanChuan.

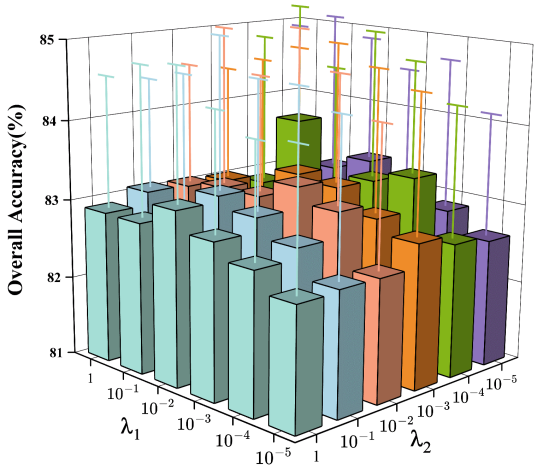


Fig. 14. OA with different λ_1 and λ_2 on Indian Pines dataset.

5) Ablation Study of the FCL: To enhance the instance discriminability of our method, we introduced a frequency-domain contrastive loss, using the original and degraded images as two perspectives for contrastive learning. This approach further strengthens the representation capability of the proposed method while narrowing the feature space between SR and classification tasks to some extent. As shown in Table V, it can be found that the OA metric is further improved, which demonstrates the significance of the proposed FCL.

F. Parameter Analysis

In this section, we analyze the effects of various configurations of key parameters, including the size of the input patch, the number of labeled samples, and loss weights.

1) Impact of Different Sizes of Input Patch: In order to analyze the impact of the size of the image patch, we conducted experiments with five different patch sizes: 8×8, 12×12, 16×16, 20×20, and 24×24, while keeping other parameters consistent. For each parameter setting, we conducted ten repeated experiments to avoid disturbance caused by initialization and reported the mean and standard deviation of the results, as shown in Fig. 12. For the Indian Pines dataset, as the patch size increases, the performance metrics initially rise and then decline. We analyze this may be because the spatial size of the Indian Pines dataset is relatively small. When the patch size is small, the spatial information captured is insufficient, and the influence of neighboring pixels is not fully utilized. As the patch size increases, performance gradually improves. However, when the patch size becomes too large,

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TABLE VII
COMPLEXITY ANALYSIS RESULTS BETWEEN OURS AND THE COMPARATIVE METHODS

		M3D-DCNN	3D-CNN	DFFN	RSSAN	SpectralFormer	SSFTT	HiT	SS-TMNet	MorphFormer	GSCViT	S2VNet	MCTGCL	Ours
IP	FLOPs(G)	0.1834	0.3780	0.1083	0.0245	0.0434	0.1411	1.1672	2.6675	0.0741	0.0839	0.0620	0.0471	14.9151
	Parameters(M)	2.8651	2.2583	0.4228	0.1102	0.3825	0.9319	51.2279	83.3265	0.1997	0.5637	0.2438	0.2234	16.2182
	OA(%)	43.60 ± 15.94	49.05 ± 2.98	64.50 ± 3.90	41.81 ± 7.89	56.57 ± 1.36	76.80 ± 1.84	53.55 ± 2.01	68.58 ± 2.99	74.80 ± 2.92	78.56 ± 2.88	78.93 ± 3.03	81.43 ± 2.91	83.82 ± 2.17
SA	FLOPs(G)	0.1862	0.3854	0.1084	0.0248	0.0445	0.1440	1.1908	2.7465	0.0756	0.0265	0.0639	0.0471	14.9164
	Parameters(M)	2.9111	2.2583	0.4234	0.1115	0.3922	0.9503	51.6526	84.5734	0.2029	0.1043	0.2504	0.2234	16.2192
	OA(%)	81.10 ± 7.78	75.37 ± 4.25	89.06 ± 1.51	77.41 ± 1.96	86.43 ± 1.50	82.78 ± 2.82	85.56 ± 2.27	90.07 ± 1.45	92.78 ± 1.14	91.01 ± 1.88	93.28 ± 1.32	93.03±0.89	94.99 ± 1.71
KSC	FLOPs(G)	0.1599	0.3320	0.1074	0.0227	0.0365	0.1240	1.0357	2.2292	0.0650	0.0246	0.0475	0.0471	14.9072
	Parameters(M)	2.0593	1.6535	0.4192	0.1002	0.3278	0.8211	48.8479	76.4029	0.1803	0.0969	0.1904	0.2231	16.2121
	OA(%)	49.51 ± 13.07	46.34 ± 11.93	85.91 ± 3.13	55.37 ± 3.54	70.37 ± 3.39	63.08 ± 5.51	64.66 ± 3.55	81.72 ± 4.75	76.60 ± 3.12	88.13 ± 2.14	84.68 ± 1.30	78.86±7.44	91.68 ± 2.65
WHU-Hi-HanChuan	FLOPs(G)	0.2525	0.5182	0.1360	0.0299	0.0671	0.1938	1.6929	4.4598	0.1021	0.0413	0.1024	0.0471	14.9393
	Parameters(M)	3.9683	3.0294	0.5309	0.1402	0.5940	1.2729	60.675	111.5696	0.2589	0.1746	0.3847	0.2236	16.2372
	OA	61.29 ± 5.79	59.29 ± 1.18	72.52 ± 2.32	61.43 ± 3.56	72.77 ± 2.28	75.88 ± 2.17	70.34 ± 5.43	77.63 ± 6.02	83.39 ± 2.13	80.02 ± 2.93	79.24 ± 2.53	85.41 ± 2.36	87.49 ± 1.94

the inclusion of excessive redundant pixels introduces noise, interfering with the decision-making of the model, which leads to a performance decline. For the Salinas, Kennedy Space Center, and WHU-Hi-HanChuan datasets, larger patch sizes yield better results, which also demonstrates the potential of our method. Since our method incorporates the SR technique, a larger patch size increases the computational cost. Therefore, a moderate patch size is ideal for balancing the trade-off between performance and computational cost. So we set the patch size to be 16×16.

2) Impact of Different Numbers of Labeled Samples: In this section, the influence of different training samples per class on our method and comparison methods is analyzed on the Indian Pines, Salinas, Kennedy Space Center, and WHU-Hi-HanChuan datasets, respectively. Fig. 13 shows the overall classification accuracy for each classifier under different numbers of labeled training samples. As with previous experiments, each setting is run ten times, and each point in the figure represents the average result. It can be observed that the performances of all methods generally improve as the numbers of labeled samples increase. Moreover, our method consistently outperforms the compared methods under all different numbers of labeled samples per class. Specifically, when fewer samples are used, our proposed method has a significant advantage over other classifiers. This may be attributed to our multitask framework, which enriches feature diversity and mitigates the adverse impact of limited labeled samples during training to some extent.

3) Loss Analysis: To evaluate the influence of different weight loss combinations, we visualize the classification performance under various settings. Each parameter combination is trained ten times, and the results are reported as shown in Fig. 14. It can be found that the OA metric increases when λ_1 increases, which demonstrates that the introduction of the SR branch leads to performance gains in classification. However, excessive constraints on the SR task may weaken the focus on the primary classification objective, leading to suboptimal classification performance. In addition, since the contrastive loss can align the feature spaces of the SR and classification tasks, the best λ_2 value is influenced by the value of λ_1 when the classification loss weight is fixed. The OA metric achieves the maximum when $\lambda_1 = 0.1$ and λ_2=1e^{-4}. So we chose this setting.

G. Analysis of the Computational Complexity

In this study, we assessed the model complexity of various methods on the Indian Pines dataset, including floating-point operations per second (FLOPs) and parameters, as shown in Table VII. It can be observed that integrating CNN and transformer architectures often leads to increased parameters and computational complexity, like HiT [26] and SS-TMNet [27]. However, compared to these hybrid architecture-based methods, our proposed approach significantly reduces the number of parameters while maintaining competitive performance. It is essential to acknowledge that there is room for improvement in efficiency, which constitutes a limitation of our method. However, considering the superior performance achieved by our method, the associated computational burden remains acceptable.

V. CONCLUSION

In this study, we integrate the SR and classification tasks into a unified framework, aiming to bridge low-level and high-level vision tasks through a frequency-domain perspective. Specifically, we construct a hybrid architecture based on transformer and CNN to learn a unified representation across different tasks. To fully exploit the advantages of both branches, we propose an FIM to achieve more powerful feature representations. Furthermore, to leverage the internal information from the SR task for assisting classification, we design a CTIM to establish effective inter-task connections. Moreover, to enhance instance discriminability, we introduce an FCL strategy by taking the original sample and its low-resolution version as two contrastive views. Finally, extensive experiments on four benchmark datasets demonstrate the effectiveness and generalization ability of the proposed method. The experimental results demonstrate that the proposed method outperforms other SOTA methods.

In our future work, we will explore optimizing the computational complexity of the multitask framework to enhance its efficiency. In addition, we will further investigate the mutual benefits between low-level and high-level vision tasks, aiming to improve the performance of the model on both tasks.

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